

2.1.7 Using Set Notation with Quantifiers

Sometimes we restrict the domain of a quantified statement explicitly by making use of a particular notation. For example, $\forall x \in S(P(x))$ denotes the universal quantification of $P(x)$ over all elements in the set S . In other words, $\forall x \in S(P(x))$ is shorthand for $\forall x(x \in S \rightarrow P(x))$. Similarly, $\exists x \in S(P(x))$ denotes the existential quantification of $P(x)$ over all elements in S . That is, $\exists x \in S(P(x))$ is shorthand for $\exists x(x \in S \wedge P(x))$.

EXAMPLE 22 What do the statements $\forall x \in \mathbf{R} (x^2 \geq 0)$ and $\exists x \in \mathbf{Z} (x^2 = 1)$ mean?

Solution: The statement $\forall x \in \mathbf{R}(x^2 \geq 0)$ states that for every real number x , $x^2 \geq 0$. This statement can be expressed as “The square of every real number is nonnegative.” This is a true statement.

The statement $\exists x \in \mathbf{Z}(x^2 = 1)$ states that there exists an integer x such that $x^2 = 1$. This statement can be expressed as “There is an integer whose square is 1.” This is also a true statement because $x = 1$ is such an integer (as is -1). ◀

2.1.8 Truth Sets and Quantifiers

We will now tie together concepts from set theory and from predicate logic. Given a predicate P , and a domain D , we define the **truth set** of P to be the set of elements x in D for which $P(x)$ is true. The truth set of $P(x)$ is denoted by $\{x \in D \mid P(x)\}$.

EXAMPLE 23 What are the truth sets of the predicates $P(x)$, $Q(x)$, and $R(x)$, where the domain is the set of integers and $P(x)$ is “ $|x| = 1$,” $Q(x)$ is “ $x^2 = 2$,” and $R(x)$ is “ $|x| = x$.”

Solution: The truth set of P , $\{x \in \mathbf{Z} \mid |x| = 1\}$, is the set of integers for which $|x| = 1$. Because $|x| = 1$ when $x = 1$ or $x = -1$, and for no other integers x , we see that the truth set of P is the set $\{-1, 1\}$.

The truth set of Q , $\{x \in \mathbf{Z} \mid x^2 = 2\}$, is the set of integers for which $x^2 = 2$. This is the empty set because there are no integers x for which $x^2 = 2$.

The truth set of R , $\{x \in \mathbf{Z} \mid |x| = x\}$, is the set of integers for which $|x| = x$. Because $|x| = x$ if and only if $x \geq 0$, it follows that the truth set of R is $\mathbf{N}$, the set of nonnegative integers. ◀

Note that $\forall x P(x)$ is true over the domain U if and only if the truth set of P is the set U . Likewise, $\exists x P(x)$ is true over the domain U if and only if the truth set of P is nonempty.

Exercises

- List the members of these sets.
 - $\{x \mid x \text{ is a real number such that } x^2 = 1\}$
 - $\{x \mid x \text{ is a positive integer less than } 12\}$
 - $\{x \mid x \text{ is the square of an integer and } x < 100\}$
 - $\{x \mid x \text{ is an integer such that } x^2 = 2\}$
- Use set builder notation to give a description of each of these sets.
 - $\{0, 3, 6, 9, 12\}$
 - $\{-3, -2, -1, 0, 1, 2, 3\}$
 - $\{m, n, o, p\}$
- Which of the intervals $(0, 5)$, $(0, 5]$, $[0, 5)$, $[0, 5]$, $(1, 4]$, $[2, 3]$, $(2, 3)$ contains
 - 0?
 - 1?
 - 2?
 - 3?
 - 4?
 - 5?
- For each of these intervals, list all its elements or explain why it is empty.
 - $[a, a]$
 - $[a, a)$
 - $(a, a]$
 - (a, a)
 - (a, b) , where $a > b$
 - $[a, b]$, where $a > b$

5. For each of these pairs of sets, determine whether the first is a subset of the second, the second is a subset of the first, or neither is a subset of the other.
 - a) the set of airline flights from New York to New Delhi, the set of nonstop airline flights from New York to New Delhi
 - b) the set of people who speak English, the set of people who speak Chinese
 - c) the set of flying squirrels, the set of living creatures that can fly
6. For each of these pairs of sets, determine whether the first is a subset of the second, the second is a subset of the first, or neither is a subset of the other.
 - a) the set of people who speak English, the set of people who speak English with an Australian accent
 - b) the set of fruits, the set of citrus fruits
 - c) the set of students studying discrete mathematics, the set of students studying data structures
7. Determine whether each of these pairs of sets are equal.
 - a) $\{1, 3, 3, 3, 5, 5, 5, 5\}$, $\{5, 3, 1\}$
 - b) $\{\{1\}\}$, $\{1, \{1\}\}$ c) $\emptyset$, $\{\emptyset\}$
8. Suppose that $A = \{2, 4, 6\}$, $B = \{2, 6\}$, $C = \{4, 6\}$, and $D = \{4, 6, 8\}$. Determine which of these sets are subsets of which other of these sets.
9. For each of the following sets, determine whether 2 is an element of that set.
 - a) $\{x \in \mathbf{R} \mid x \text{ is an integer greater than } 1\}$
 - b) $\{x \in \mathbf{R} \mid x \text{ is the square of an integer}\}$
 - c) $\{2, \{2\}\}$ d) $\{\{2\}, \{\{2\}\}\}$
 - e) $\{\{2\}, \{2, \{2\}\}\}$ f) $\{\{\{2\}\}\}$
10. For each of the sets in Exercise 9, determine whether $\{2\}$ is an element of that set.
11. Determine whether each of these statements is true or false.
 - a) $0 \in \emptyset$ b) $\emptyset \in \{0\}$
 - c) $\{0\} \subset \emptyset$ d) $\emptyset \subset \{0\}$
 - e) $\{0\} \in \{0\}$ f) $\{0\} \subset \{0\}$
 - g) $\{\emptyset\} \subseteq \{\emptyset\}$
12. Determine whether these statements are true or false.
 - a) $\emptyset \in \{\emptyset\}$ b) $\emptyset \in \{\emptyset, \{\emptyset\}\}$
 - c) $\{\emptyset\} \in \{\emptyset\}$ d) $\{\emptyset\} \in \{\{\emptyset\}\}$
 - e) $\{\emptyset\} \subset \{\emptyset, \{\emptyset\}\}$ f) $\{\{\emptyset\}\} \subset \{\emptyset, \{\emptyset\}\}$
 - g) $\{\{\emptyset\}\} \subset \{\{\emptyset\}, \{\emptyset\}\}$
13. Determine whether each of these statements is true or false.
 - a) $x \in \{x\}$ b) $\{x\} \subseteq \{x\}$ c) $\{x\} \in \{x\}$
 - d) $\{x\} \in \{\{x\}\}$ e) $\emptyset \subseteq \{x\}$ f) $\emptyset \in \{x\}$
14. Use a Venn diagram to illustrate the subset of odd integers in the set of all positive integers not exceeding 10.
15. Use a Venn diagram to illustrate the set of all months of the year whose names do not contain the letter R in the set of all months of the year.
16. Use a Venn diagram to illustrate the relationship $A \subseteq B$ and $B \subseteq C$.
17. Use a Venn diagram to illustrate the relationships $A \subset B$ and $B \subset C$.
18. Use a Venn diagram to illustrate the relationships $A \subset B$ and $A \subset C$.
19. Suppose that A , B , and C are sets such that $A \subseteq B$ and $B \subseteq C$. Show that $A \subseteq C$.
20. Find two sets A and B such that $A \in B$ and $A \subseteq B$.
21. What is the cardinality of each of these sets?
 - a) $\{a\}$ b) $\{\{a\}\}$
 - c) $\{a, \{a\}\}$ d) $\{a, \{a\}, \{a, \{a\}\}\}$
22. What is the cardinality of each of these sets?
 - a) $\emptyset$ b) $\{\emptyset\}$
 - c) $\{\emptyset, \{\emptyset\}\}$ d) $\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$
23. Find the power set of each of these sets, where a and b are distinct elements.
 - a) $\{a\}$ b) $\{a, b\}$ c) $\{\emptyset, \{\emptyset\}\}$
24. Can you conclude that $A = B$ if A and B are two sets with the same power set?
25. How many elements does each of these sets have where a and b are distinct elements?
 - a) $\mathcal{P}(\{a, b, \{a, b\}\})$
 - b) $\mathcal{P}(\{\emptyset, a, \{a\}, \{\{a\}\}\})$
 - c) $\mathcal{P}(\mathcal{P}(\emptyset))$
26. Determine whether each of these sets is the power set of a set, where a and b are distinct elements.
 - a) $\emptyset$ b) $\{\emptyset, \{a\}\}$
 - c) $\{\emptyset, \{a\}, \{\emptyset, a\}\}$ d) $\{\emptyset, \{a\}, \{b\}, \{a, b\}\}$
27. Prove that $\mathcal{P}(A) \subseteq \mathcal{P}(B)$ if and only if $A \subseteq B$.
28. Show that if $A \subseteq C$ and $B \subseteq D$, then $A \times B \subseteq C \times D$.
29. Let $A = \{a, b, c, d\}$ and $B = \{y, z\}$. Find
 - a) $A \times B$. b) $B \times A$.
30. What is the Cartesian product $A \times B$, where A is the set of courses offered by the mathematics department at a university and B is the set of mathematics professors at this university? Give an example of how this Cartesian product can be used.
31. What is the Cartesian product $A \times B \times C$, where A is the set of all airlines and B and C are both the set of all cities in the United States? Give an example of how this Cartesian product can be used.
32. Suppose that $A \times B = \emptyset$, where A and B are sets. What can you conclude?
33. Let A be a set. Show that $\emptyset \times A = A \times \emptyset = \emptyset$.
34. Let $A = \{a, b, c\}$, $B = \{x, y\}$, and $C = \{0, 1\}$. Find
 - a) $A \times B \times C$. b) $C \times B \times A$.
 - c) $C \times A \times B$. d) $B \times B \times B$.
35. Find A^2 if
 - a) $A = \{0, 1, 3\}$. b) $A = \{1, 2, a, b\}$.
36. Find A^3 if
 - a) $A = \{a\}$. b) $A = \{0, a\}$.
37. How many different elements does $A \times B$ have if A has m elements and B has n elements?
38. How many different elements does $A \times B \times C$ have if A has m elements, B has n elements, and C has p elements?

39. How many different elements does A^n have when A has m elements and n is a positive integer?
40. Show that $A \times B \neq B \times A$, when A and B are nonempty, unless $A = B$.
41. Explain why $A \times B \times C$ and $(A \times B) \times C$ are not the same.
42. Explain why $(A \times B) \times (C \times D)$ and $A \times (B \times C) \times D$ are not the same.
43. Prove or disprove that if A and B are sets, then $\mathcal{P}(A \times B) = \mathcal{P}(A) \times \mathcal{P}(B)$.
44. Prove or disprove that if A , B , and C are nonempty sets and $A \times B = A \times C$, then $B = C$.
45. Translate each of these quantifications into English and determine its truth value.
- a) $\forall x \in \mathbf{R} (x^2 \neq -1)$ b) $\exists x \in \mathbf{Z} (x^2 = 2)$
 c) $\forall x \in \mathbf{Z} (x^2 > 0)$ d) $\exists x \in \mathbf{R} (x^2 = x)$
46. Translate each of these quantifications into English and determine its truth value.
- a) $\exists x \in \mathbf{R} (x^3 = -1)$ b) $\exists x \in \mathbf{Z} (x + 1 > x)$
 c) $\forall x \in \mathbf{Z} (x - 1 \in \mathbf{Z})$ d) $\forall x \in \mathbf{Z} (x^2 \in \mathbf{Z})$
47. Find the truth set of each of these predicates where the domain is the set of integers.
- a) $P(x): x^2 < 3$ b) $Q(x): x^2 > x$
 c) $R(x): 2x + 1 = 0$
48. Find the truth set of each of these predicates where the domain is the set of integers.
- a) $P(x): x^3 \geq 1$ b) $Q(x): x^2 = 2$
 c) $R(x): x < x^2$
- *49. The defining property of an ordered pair is that two ordered pairs are equal if and only if their first elements are equal and their second elements are equal. Surprisingly, instead of taking the ordered pair as a primitive concept, we can construct ordered pairs using basic notions from set theory. Show that if we define the ordered pair (a, b) to be $\{\{a\}, \{a, b\}\}$, then $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$. [Hint: First show that $\{\{a\}, \{a, b\}\} = \{\{c\}, \{c, d\}\}$ if and only if $a = c$ and $b = d$.]
- *50. This exercise presents **Russell's paradox**. Let S be the set that contains a set x if the set x does not belong to itself, so that $S = \{x \mid x \notin x\}$.
- a) Show the assumption that S is a member of S leads to a contradiction.
 b) Show the assumption that S is not a member of S leads to a contradiction.
- By parts (a) and (b) it follows that the set S cannot be defined as it was. This paradox can be avoided by restricting the types of elements that sets can have.
- *51. Describe a procedure for listing all the subsets of a finite set.

2.2 Set Operations

2.2.1 Introduction



Two, or more, sets can be combined in many different ways. For instance, starting with the set of mathematics majors at your school and the set of computer science majors at your school, we can form the set of students who are mathematics majors or computer science majors, the set of students who are joint majors in mathematics and computer science, the set of all students not majoring in mathematics, and so on.

Definition 1

Let A and B be sets. The *union* of the sets A and B , denoted by $A \cup B$, is the set that contains those elements that are either in A or in B , or in both.

An element x belongs to the union of the sets A and B if and only if x belongs to A or x belongs to B . This tells us that

$$A \cup B = \{x \mid x \in A \vee x \in B\}.$$

The Venn diagram shown in Figure 1 represents the union of two sets A and B . The area that represents $A \cup B$ is the shaded area within either the circle representing A or the circle representing B .

We will give some examples of the union of sets.

EXAMPLE 1 The union of the sets $\{1, 3, 5\}$ and $\{1, 2, 3\}$ is the set $\{1, 2, 3, 5\}$; that is, $\{1, 3, 5\} \cup \{1, 2, 3\} = \{1, 2, 3, 5\}$. 

$$\begin{aligned}
 P - Q &= \{\max(4 - 3, 0) \cdot a, \max(1 - 4, 0) \cdot b, \max(3 - 0, 0) \cdot c, \max(0 - 2, 0) \cdot d\} \\
 &= \{1 \cdot a, 0 \cdot b, 3 \cdot c, 0 \cdot d\} = \{1 \cdot a, 3 \cdot c\}, \text{ and}
 \end{aligned}$$

$$\begin{aligned}
 P + Q &= \{(4 + 3) \cdot a, (1 + 4) \cdot b, (3 + 0) \cdot c, (0 + 2) \cdot d\} \\
 &= \{7 \cdot a, 5 \cdot b, 3 \cdot c, 2 \cdot d\}.
 \end{aligned}$$

Exercises

- Let A be the set of students who live within one mile of school and let B be the set of students who walk to classes. Describe the students in each of these sets.
 - $A \cap B$
 - $A \cup B$
 - $A - B$
 - $B - A$
 - Suppose that A is the set of sophomores at your school and B is the set of students in discrete mathematics at your school. Express each of these sets in terms of A and B .
 - the set of sophomores taking discrete mathematics in your school
 - the set of sophomores at your school who are not taking discrete mathematics
 - the set of students at your school who either are sophomores or are taking discrete mathematics
 - the set of students at your school who either are not sophomores or are not taking discrete mathematics
 - Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{0, 3, 6\}$. Find
 - $A \cup B$.
 - $A \cap B$.
 - $A - B$.
 - $B - A$.
 - Let $A = \{a, b, c, d, e\}$ and $B = \{a, b, c, d, e, f, g, h\}$. Find
 - $A \cup B$.
 - $A \cap B$.
 - $A - B$.
 - $B - A$.
- In Exercises 5–10 assume that A is a subset of some underlying universal set U .
- Prove the complementation law in Table 1 by showing that $\overline{\overline{A}} = A$.
 - Prove the identity laws in Table 1 by showing that
 - $A \cup \emptyset = A$.
 - $A \cap U = A$.
 - Prove the domination laws in Table 1 by showing that
 - $A \cup U = U$.
 - $A \cap \emptyset = \emptyset$.
 - Prove the idempotent laws in Table 1 by showing that
 - $A \cup A = A$.
 - $A \cap A = A$.
 - Prove the complement laws in Table 1 by showing that
 - $A \cup \overline{A} = U$.
 - $A \cap \overline{A} = \emptyset$.
 - Show that
 - $A - \emptyset = A$.
 - $\emptyset - A = \emptyset$.
 - Let A and B be sets. Prove the commutative laws from Table 1 by showing that
 - $A \cup B = B \cup A$.
 - $A \cap B = B \cap A$.
 - Prove the first absorption law from Table 1 by showing that if A and B are sets, then $A \cup (A \cap B) = A$.
 - Prove the second absorption law from Table 1 by showing that if A and B are sets, then $A \cap (A \cup B) = A$.
 - Find the sets A and B if $A - B = \{1, 5, 7, 8\}$, $B - A = \{2, 10\}$, and $A \cap B = \{3, 6, 9\}$.
 - Prove the second De Morgan law in Table 1 by showing that if A and B are sets, then $\overline{A \cup B} = \overline{A} \cap \overline{B}$
 - by showing each side is a subset of the other side.
 - using a membership table.
 - Let A and B be sets. Show that
 - $(A \cap B) \subseteq A$.
 - $A \subseteq (A \cup B)$.
 - $A - B \subseteq A$.
 - $A \cap (B - A) = \emptyset$.
 - $A \cup (B - A) = A \cup B$.
 - Show that if A and B are sets in a universe U then $A \subseteq B$ if and only if $\overline{A} \cup B = U$.
 - Given sets A and B in a universe U , draw the Venn diagrams of each of these sets.
 - $A \rightarrow B = \{x \in U \mid x \in A \rightarrow x \in B\}$
 - $A \leftrightarrow B = \{x \in U \mid x \in A \leftrightarrow x \in B\}$
 - Show that if A , B , and C are sets, then $\overline{A \cap B \cap C} = \overline{A} \cup \overline{B} \cup \overline{C}$
 - by showing each side is a subset of the other side.
 - using a membership table.
 - Let A , B , and C be sets. Show that
 - $(A \cup B) \subseteq (A \cup B \cup C)$.
 - $(A \cap B \cap C) \subseteq (A \cap B)$.
 - $(A - B) - C \subseteq A - C$.
 - $(A - C) \cap (C - B) = \emptyset$.
 - $(B - A) \cup (C - A) = (B \cup C) - A$.
 - Show that if A and B are sets, then
 - $A - B = A \cap \overline{B}$.
 - $(A \cap B) \cup (A \cap \overline{B}) = A$.
 - Show that if A and B are sets with $A \subseteq B$, then
 - $A \cup B = B$.
 - $A \cap B = A$.
 - Prove the first associative law from Table 1 by showing that if A , B , and C are sets, then $A \cup (B \cap C) = (A \cup B) \cap C$.
 - Prove the second associative law from Table 1 by showing that if A , B , and C are sets, then $A \cap (B \cup C) = (A \cap B) \cup C$.
 - Prove the first distributive law from Table 1 by showing that if A , B , and C are sets, then $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

26. Let A , B , and C be sets. Show that $(A - B) - C = (A - C) - (B - C)$.
27. Let $A = \{0, 2, 4, 6, 8, 10\}$, $B = \{0, 1, 2, 3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7, 8, 9, 10\}$. Find
- $A \cap B \cap C$.
 - $A \cup B \cup C$.
 - $(A \cup B) \cap C$.
 - $(A \cap B) \cup C$.
28. Draw the Venn diagrams for each of these combinations of the sets A , B , and C .
- $A \cap (B \cup C)$
 - $\overline{A} \cap \overline{B} \cap \overline{C}$
 - $(A - B) \cup (A - C) \cup (B - C)$
29. Draw the Venn diagrams for each of these combinations of the sets A , B , and C .
- $A \cap (B - C)$
 - $(A \cap B) \cup (A \cap C)$
 - $(A \cap \overline{B}) \cup (A \cap \overline{C})$
30. Draw the Venn diagrams for each of these combinations of the sets A , B , C , and D .
- $(A \cap B) \cup (C \cap D)$
 - $\overline{A} \cup \overline{B} \cup \overline{C} \cup \overline{D}$
 - $A - (B \cap C \cap D)$
31. What can you say about the sets A and B if we know that
- $A \cup B = A$?
 - $A \cap B = A$?
 - $A - B = A$?
 - $A \cap B = B \cap A$?
 - $A - B = B - A$?
32. Can you conclude that $A = B$ if A , B , and C are sets such that
- $A \cup C = B \cup C$?
 - $A \cap C = B \cap C$?
 - $A \cup C = B \cup C$ and $A \cap C = B \cap C$?
33. Let A and B be subsets of a universal set U . Show that $A \subseteq B$ if and only if $\overline{B} \subseteq \overline{A}$.
34. Let A , B , and C be sets. Use the identity $A - B = A \cap \overline{B}$, which holds for any sets A and B , and the identities from Table 1 to show that $(A - B) \cap (B - C) \cap (A - C) = \emptyset$.
35. Let A , B , and C be sets. Use the identities in Table 1 to show that $(A \cup B) \cap (B \cup C) \cap (A \cup C) = A \cap B \cap C$.
36. Prove or disprove that for all sets A , B , and C , we have
- $A \times (B \cup C) = (A \times B) \cup (A \times C)$.
 - $A \times (B \cap C) = (A \times B) \cap (A \times C)$.
37. Prove or disprove that for all sets A , B , and C , we have
- $A \times (B - C) = (A \times B) - (A \times C)$.
 - $A \times (B \cup C) = A \times (B \cup C)$.
- The **symmetric difference** of A and B , denoted by $A \oplus B$, is the set containing those elements in either A or B , but not in both A and B .
38. Find the symmetric difference of $\{1, 3, 5\}$ and $\{1, 2, 3\}$.
39. Find the symmetric difference of the set of computer science majors at a school and the set of mathematics majors at this school.
40. Draw a Venn diagram for the symmetric difference of the sets A and B .
41. Show that $A \oplus B = (A \cup B) - (A \cap B)$.
42. Show that $A \oplus B = (A - B) \cup (B - A)$.
43. Show that if A is a subset of a universal set U , then
- $A \oplus A = \emptyset$.
 - $A \oplus \emptyset = A$.
 - $A \oplus U = \overline{A}$.
 - $A \oplus \overline{A} = U$.
44. Show that if A and B are sets, then
- $A \oplus B = B \oplus A$.
 - $(A \oplus B) \oplus B = A$.
45. What can you say about the sets A and B if $A \oplus B = A$?
- *46. Determine whether the symmetric difference is associative; that is, if A , B , and C are sets, does it follow that $A \oplus (B \oplus C) = (A \oplus B) \oplus C$?
- *47. Suppose that A , B , and C are sets such that $A \oplus C = B \oplus C$. Must it be the case that $A = B$?
48. If A , B , C , and D are sets, does it follow that $(A \oplus B) \oplus (C \oplus D) = (A \oplus C) \oplus (B \oplus D)$?
49. If A , B , C , and D are sets, does it follow that $(A \oplus B) \oplus (C \oplus D) = (A \oplus D) \oplus (B \oplus C)$?
50. Show that if A and B are finite sets, then $A \cup B$ is a finite set.
51. Show that if A is an infinite set, then whenever B is a set, $A \cup B$ is also an infinite set.
- *52. Show that if A , B , and C are sets, then
- $$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$
- (This is a special case of the inclusion-exclusion principle, which will be studied in Chapter 8.)
53. Let $A_i = \{1, 2, 3, \dots, i\}$ for $i = 1, 2, 3, \dots$. Find
- $\bigcup_{i=1}^n A_i$.
 - $\bigcap_{i=1}^n A_i$.
54. Let $A_i = \{\dots, -2, -1, 0, 1, \dots, i\}$. Find
- $\bigcup_{i=1}^n A_i$.
 - $\bigcap_{i=1}^n A_i$.
55. Let A_i be the set of all nonempty bit strings (that is, bit strings of length at least one) of length not exceeding i . Find
- $\bigcup_{i=1}^n A_i$.
 - $\bigcap_{i=1}^n A_i$.
56. Find $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$ if for every positive integer i ,
- $A_i = \{i, i+1, i+2, \dots\}$.
 - $A_i = \{0, i\}$.
 - $A_i = (0, i)$, that is, the set of real numbers x with $0 < x < i$.
 - $A_i = (i, \infty)$, that is, the set of real numbers x with $x > i$.
57. Find $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$ if for every positive integer i ,
- $A_i = \{-i, -i+1, \dots, -1, 0, 1, \dots, i-1, i\}$.
 - $A_i = \{-i, i\}$.
 - $A_i = [-i, i]$, that is, the set of real numbers x with $-i \leq x \leq i$.
 - $A_i = [i, \infty)$, that is, the set of real numbers x with $x \geq i$.
58. Suppose that the universal set is $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Express each of these sets with bit strings where the i th bit in the string is 1 if i is in the set and 0 otherwise.
- $\{3, 4, 5\}$
 - $\{1, 3, 6, 10\}$
 - $\{2, 3, 4, 7, 8, 9\}$

59. Using the same universal set as in the last exercise, find the set specified by each of these bit strings.
- 11 1100 1111
 - 01 0111 1000
 - 10 0000 0001
60. What subsets of a finite universal set do these bit strings represent?
- the string with all zeros
 - the string with all ones
61. What is the bit string corresponding to the difference of two sets?
62. What is the bit string corresponding to the symmetric difference of two sets?
63. Show how bitwise operations on bit strings can be used to find these combinations of $A = \{a, b, c, d, e\}$, $B = \{b, c, d, g, p, t, v\}$, $C = \{c, e, i, o, u, x, y, z\}$, and $D = \{d, e, h, i, n, o, t, u, x, y\}$.
- $A \cup B$
 - $A \cap B$
 - $(A \cup D) \cap (B \cup C)$
 - $A \cup B \cup C \cup D$
64. How can the union and intersection of n sets that all are subsets of the universal set U be found using bit strings?
- The **successor** of the set A is the set $A \cup \{A\}$.
65. Find the successors of the following sets.
- $\{1, 2, 3\}$
 - $\emptyset$
 - $\{\emptyset\}$
 - $\{\emptyset, \{\emptyset\}\}$
66. How many elements does the successor of a set with n elements have?
67. Let A and B be the multisets $\{3 \cdot a, 2 \cdot b, 1 \cdot c\}$ and $\{2 \cdot a, 3 \cdot b, 4 \cdot d\}$, respectively. Find
- $A \cup B$.
 - $A \cap B$.
 - $A - B$.
 - $B - A$.
 - $A + B$.
68. Assume that $a \in A$, where A is a set. Which of these statements are true and which are false, where all sets shown are ordinary sets, and not multisets. Explain each answer.
- $\{a, a\} \cup \{a, a, a\} = \{a, a, a, a, a\}$
 - $\{a, a\} \cup \{a, a, a\} = \{a\}$
 - $\{a, a\} \cap \{a, a, a\} = \{a, a\}$
 - $\{a, a\} \cap \{a, a, a\} = \{a\}$
 - $\{a, a, a\} - \{a, a\} = \{a\}$
69. Answer the same questions as posed in Exercise 68 where all sets are multisets, and not ordinary sets.
70. Suppose that A is the multiset that has as its elements the types of computer equipment needed by one department of a university and the multiplicities are the number of pieces of each type needed, and B is the analogous multiset for a second department of the university. For instance, A could be the multiset $\{107 \cdot \text{personal computers}, 44 \cdot \text{routers}, 6 \cdot \text{servers}\}$ and B could be the multiset $\{14 \cdot \text{personal computers}, 6 \cdot \text{routers}, 2 \cdot \text{mainframes}\}$.
- What combination of A and B represents the equipment the university should buy assuming both departments use the same equipment?
 - What combination of A and B represents the equipment that will be used by both departments if both departments use the same equipment?

- What combination of A and B represents the equipment that the second department uses, but the first department does not, if both departments use the same equipment?
- What combination of A and B represents the equipment that the university should purchase if the departments do not share equipment?

The **Jaccard similarity** $J(A, B)$ of the finite sets A and B is $J(A, B) = |A \cap B| / |A \cup B|$, with $J(\emptyset, \emptyset) = 1$. The **Jaccard distance** $d_J(A, B)$ between A and B equals $d_J(A, B) = 1 - J(A, B)$.

71. Find $J(A, B)$ and $d_J(A, B)$ for these pairs of sets.
- $A = \{1, 3, 5\}$, $B = \{2, 4, 6\}$
 - $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$
 - $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{1, 2, 3, 4, 5, 6\}$
 - $A = \{1\}$, $B = \{1, 2, 3, 4, 5, 6\}$
72. Prove that each of the properties in parts (a)–(d) holds whenever A and B are finite sets.
- $J(A, A) = 1$ and $d_J(A, A) = 0$
 - $J(A, B) = J(B, A)$ and $d_J(A, B) = d_J(B, A)$
 - $J(A, B) = 1$ and $d_J(A, B) = 0$ if and only if $A = B$
 - $0 \leq J(A, B) \leq 1$ and $0 \leq d_J(A, B) \leq 1$
- **e)** Show that if A , B , and C are sets, then $d_J(A, C) \leq d_J(A, B) + d_J(B, C)$. (This inequality is known as the **triangle inequality** and together with parts (a), (b), and (c) implies that d_J is a **metric**.)

Fuzzy sets are used in artificial intelligence. Each element in the universal set U has a **degree of membership**, which is a real number between 0 and 1 (including 0 and 1), in a fuzzy set S . The fuzzy set S is denoted by listing the elements with their degrees of membership (elements with 0 degree of membership are not listed). For instance, we write $\{0.6 \text{ Alice}, 0.9 \text{ Brian}, 0.4 \text{ Fred}, 0.1 \text{ Oscar}, 0.5 \text{ Rita}\}$ for the set F (of famous people) to indicate that Alice has a 0.6 degree of membership in F , Brian has a 0.9 degree of membership in F , Fred has a 0.4 degree of membership in F , Oscar has a 0.1 degree of membership in F , and Rita has a 0.5 degree of membership in F (so that Brian is the most famous and Oscar is the least famous of these people). Also suppose that R is the set of rich people with $R = \{0.4 \text{ Alice}, 0.8 \text{ Brian}, 0.2 \text{ Fred}, 0.9 \text{ Oscar}, 0.7 \text{ Rita}\}$.

73. The **complement** of a fuzzy set S is the set $\bar{S}$, with the degree of the membership of an element in $\bar{S}$ equal to 1 minus the degree of membership of this element in S . Find $\bar{F}$ (the fuzzy set of people who are not famous) and $\bar{R}$ (the fuzzy set of people who are not rich).
74. The **union** of two fuzzy sets S and T is the fuzzy set $S \cup T$, where the degree of membership of an element in $S \cup T$ is the maximum of the degrees of membership of this element in S and in T . Find the fuzzy set $F \cup R$ of rich or famous people.
75. The **intersection** of two fuzzy sets S and T is the fuzzy set $S \cap T$, where the degree of membership of an element in $S \cap T$ is the minimum of the degrees of membership of this element in S and in T . Find the fuzzy set $F \cap R$ of rich and famous people.

2.3.6 Partial Functions

A program designed to evaluate a function may not produce the correct value of the function for all elements in the domain of this function. For example, a program may not produce a correct value because evaluating the function may lead to an infinite loop or an overflow. Similarly, in abstract mathematics, we often want to discuss functions that are defined only for a subset of the real numbers, such as $1/x$, $\sqrt{x}$, and $\arcsin(x)$. Also, we may want to use such notions as the “youngest child” function, which is undefined for a couple having no children, or the “time of sunrise,” which is undefined for some days above the Arctic Circle. To study such situations, we use the concept of a partial function.

Definition 13

A *partial function* f from a set A to a set B is an assignment to each element a in a subset of A , called the *domain of definition* of f , of a unique element b in B . The sets A and B are called the *domain* and *codomain* of f , respectively. We say that f is *undefined* for elements in A that are not in the domain of definition of f . When the domain of definition of f equals A , we say that f is a *total function*.

Remark: We write $f : A \rightarrow B$ to denote that f is a partial function from A to B . Note that this is the same notation as is used for functions. The context in which the notation is used determines whether f is a partial function or a total function.

EXAMPLE 34

The function $f : \mathbf{Z} \rightarrow \mathbf{R}$ where $f(n) = \sqrt{n}$ is a partial function from $\mathbf{Z}$ to $\mathbf{R}$ where the domain of definition is the set of nonnegative integers. Note that f is undefined for negative integers. ◀

Exercises

- Why is f not a function from $\mathbf{R}$ to $\mathbf{R}$ if
 - $f(x) = 1/x$?
 - $f(x) = \sqrt{x}$?
 - $f(x) = \pm\sqrt{(x^2 + 1)}$?
- Determine whether f is a function from $\mathbf{Z}$ to $\mathbf{R}$ if
 - $f(n) = \pm n$.
 - $f(n) = \sqrt{n^2 + 1}$.
 - $f(n) = 1/(n^2 - 4)$.
- Determine whether f is a function from the set of all bit strings to the set of integers if
 - $f(S)$ is the position of a 0 bit in S .
 - $f(S)$ is the number of 1 bits in S .
 - $f(S)$ is the smallest integer i such that the i th bit of S is 1 and $f(S) = 0$ when S is the empty string, the string with no bits.
- Find the domain and range of these functions. Note that in each case, to find the domain, determine the set of elements assigned values by the function.
 - the function that assigns to each nonnegative integer its last digit
 - the function that assigns the next largest integer to a positive integer
 - the function that assigns to a bit string the number of one bits in the string
 - the function that assigns to a bit string the number of bits in the string
- Find the domain and range of these functions. Note that in each case, to find the domain, determine the set of elements assigned values by the function.
 - the function that assigns to each bit string the number of ones in the string minus the number of zeros in the string
 - the function that assigns to each bit string twice the number of zeros in that string
 - the function that assigns the number of bits left over when a bit string is split into bytes (which are blocks of 8 bits)
 - the function that assigns to each positive integer the largest perfect square not exceeding this integer
- Find the domain and range of these functions.
 - the function that assigns to each pair of positive integers the first integer of the pair
 - the function that assigns to each positive integer its largest decimal digit
 - the function that assigns to a bit string the number of ones minus the number of zeros in the string
 - the function that assigns to each positive integer the largest integer not exceeding the square root of the integer
 - the function that assigns to a bit string the longest string of ones in the string

7. Find the domain and range of these functions.
 - a) the function that assigns to each pair of positive integers the maximum of these two integers
 - b) the function that assigns to each positive integer the number of the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 that do not appear as decimal digits of the integer
 - c) the function that assigns to a bit string the number of times the block 11 appears
 - d) the function that assigns to a bit string the numerical position of the first 1 in the string and that assigns the value 0 to a bit string consisting of all 0s
8. Find these values.
 - a) $\lfloor 1.1 \rfloor$
 - b) $\lceil 1.1 \rceil$
 - c) $\lfloor -0.1 \rfloor$
 - d) $\lceil -0.1 \rceil$
 - e) $\lfloor 2.99 \rfloor$
 - f) $\lceil -2.99 \rceil$
 - g) $\lfloor \frac{1}{2} \rfloor + \lceil \frac{1}{2} \rceil$
 - h) $\lceil \lfloor \frac{1}{2} \rfloor + \lceil \frac{1}{2} \rceil + \frac{1}{2} \rceil$
9. Find these values.
 - a) $\lceil \frac{3}{4} \rceil$
 - b) $\lfloor \frac{7}{8} \rfloor$
 - c) $\lceil -\frac{3}{4} \rceil$
 - d) $\lfloor -\frac{7}{8} \rfloor$
 - e) $\lceil 3 \rceil$
 - f) $\lfloor -1 \rfloor$
 - g) $\lfloor \frac{1}{2} + \lceil \frac{3}{2} \rceil \rfloor$
 - h) $\lfloor \frac{1}{2} \cdot \lfloor \frac{5}{2} \rfloor \rfloor$
10. Determine whether each of these functions from $\{a, b, c, d\}$ to itself is one-to-one.
 - a) $f(a) = b, f(b) = a, f(c) = c, f(d) = d$
 - b) $f(a) = b, f(b) = b, f(c) = d, f(d) = c$
 - c) $f(a) = d, f(b) = b, f(c) = c, f(d) = d$
11. Which functions in Exercise 10 are onto?
12. Determine whether each of these functions from $\mathbf{Z}$ to $\mathbf{Z}$ is one-to-one.
 - a) $f(n) = n - 1$
 - b) $f(n) = n^2 + 1$
 - c) $f(n) = n^3$
 - d) $f(n) = \lfloor n/2 \rfloor$
13. Which functions in Exercise 12 are onto?
14. Determine whether $f: \mathbf{Z} \times \mathbf{Z} \rightarrow \mathbf{Z}$ is onto if
 - a) $f(m, n) = 2m - n$.
 - b) $f(m, n) = m^2 - n^2$.
 - c) $f(m, n) = m + n + 1$.
 - d) $f(m, n) = |m| - |n|$.
 - e) $f(m, n) = m^2 - 4$.
15. Determine whether the function $f: \mathbf{Z} \times \mathbf{Z} \rightarrow \mathbf{Z}$ is onto if
 - a) $f(m, n) = m + n$.
 - b) $f(m, n) = m^2 + n^2$.
 - c) $f(m, n) = m$.
 - d) $f(m, n) = |n|$.
 - e) $f(m, n) = m - n$.
16. Consider these functions from the set of students in a discrete mathematics class. Under what conditions is the function one-to-one if it assigns to a student his or her
 - a) mobile phone number.
 - b) student identification number.
 - c) final grade in the class.
 - d) home town.
17. Consider these functions from the set of teachers in a school. Under what conditions is the function one-to-one if it assigns to a teacher his or her
 - a) office.
 - b) assigned bus to chaperone in a group of buses taking students on a field trip.
 - c) salary.
 - d) social security number.
18. Specify a codomain for each of the functions in Exercise 16. Under what conditions is each of these functions with the codomain you specified onto?
19. Specify a codomain for each of the functions in Exercise 17. Under what conditions is each of the functions with the codomain you specified onto?
20. Give an example of a function from $\mathbf{N}$ to $\mathbf{N}$ that is
 - a) one-to-one but not onto.
 - b) onto but not one-to-one.
 - c) both onto and one-to-one (but different from the identity function).
 - d) neither one-to-one nor onto.
21. Give an explicit formula for a function from the set of integers to the set of positive integers that is
 - a) one-to-one, but not onto.
 - b) onto, but not one-to-one.
 - c) one-to-one and onto.
 - d) neither one-to-one nor onto.
22. Determine whether each of these functions is a bijection from $\mathbf{R}$ to $\mathbf{R}$.
 - a) $f(x) = -3x + 4$
 - b) $f(x) = -3x^2 + 7$
 - c) $f(x) = (x + 1)/(x + 2)$
 - d) $f(x) = x^5 + 1$
23. Determine whether each of these functions is a bijection from $\mathbf{R}$ to $\mathbf{R}$.
 - a) $f(x) = 2x + 1$
 - b) $f(x) = x^2 + 1$
 - c) $f(x) = x^3$
 - d) $f(x) = (x^2 + 1)/(x^2 + 2)$
24. Let $f: \mathbf{R} \rightarrow \mathbf{R}$ and let $f(x) > 0$ for all $x \in \mathbf{R}$. Show that $f(x)$ is strictly increasing if and only if the function $g(x) = 1/f(x)$ is strictly decreasing.
25. Let $f: \mathbf{R} \rightarrow \mathbf{R}$ and let $f(x) > 0$ for all $x \in \mathbf{R}$. Show that $f(x)$ is strictly decreasing if and only if the function $g(x) = 1/f(x)$ is strictly increasing.
26. a) Prove that a strictly increasing function from $\mathbf{R}$ to itself is one-to-one.
b) Give an example of an increasing function from $\mathbf{R}$ to itself that is not one-to-one.
27. a) Prove that a strictly decreasing function from $\mathbf{R}$ to itself is one-to-one.
b) Give an example of a decreasing function from $\mathbf{R}$ to itself that is not one-to-one.
28. Show that the function $f(x) = e^x$ from the set of real numbers to the set of real numbers is not invertible, but if the codomain is restricted to the set of positive real numbers, the resulting function is invertible.

29. Show that the function $f(x) = |x|$ from the set of real numbers to the set of nonnegative real numbers is not invertible, but if the domain is restricted to the set of nonnegative real numbers, the resulting function is invertible.
30. Let $S = \{-1, 0, 2, 4, 7\}$. Find $f(S)$ if
- $f(x) = 1$.
 - $f(x) = 2x + 1$.
 - $f(x) = \lfloor x/5 \rfloor$.
 - $f(x) = \lfloor (x^2 + 1)/3 \rfloor$.
31. Let $f(x) = \lfloor x^2/3 \rfloor$. Find $f(S)$ if
- $S = \{-2, -1, 0, 1, 2, 3\}$.
 - $S = \{0, 1, 2, 3, 4, 5\}$.
 - $S = \{1, 5, 7, 11\}$.
 - $S = \{2, 6, 10, 14\}$.
32. Let $f(x) = 2x$ where the domain is the set of real numbers. What is
- $f(\mathbf{Z})$?
 - $f(\mathbf{N})$?
 - $f(\mathbf{R})$?
33. Suppose that g is a function from A to B and f is a function from B to C .
- Show that if both f and g are one-to-one functions, then $f \circ g$ is also one-to-one.
 - Show that if both f and g are onto functions, then $f \circ g$ is also onto.
34. Suppose that g is a function from A to B and f is a function from B to C . Prove each of these statements.
- If $f \circ g$ is onto, then f must also be onto.
 - If $f \circ g$ is one-to-one, then g must also be one-to-one.
 - If $f \circ g$ is a bijection, then g is onto if and only if f is one-to-one.
35. Find an example of functions f and g such that $f \circ g$ is a bijection, but g is not onto and f is not one-to-one.
- *36. If f and $f \circ g$ are one-to-one, does it follow that g is one-to-one? Justify your answer.
- *37. If f and $f \circ g$ are onto, does it follow that g is onto? Justify your answer.
38. Find $f \circ g$ and $g \circ f$, where $f(x) = x^2 + 1$ and $g(x) = x + 2$, are functions from $\mathbf{R}$ to $\mathbf{R}$.
39. Find $f + g$ and fg for the functions f and g given in Exercise 36.
40. Let $f(x) = ax + b$ and $g(x) = cx + d$, where a, b, c , and d are constants. Determine necessary and sufficient conditions on the constants a, b, c , and d so that $f \circ g = g \circ f$.
41. Show that the function $f(x) = ax + b$ from $\mathbf{R}$ to $\mathbf{R}$, where a and b are constants with $a \neq 0$ is invertible, and find the inverse of f .
42. Let f be a function from the set A to the set B . Let S and T be subsets of A . Show that
- $f(S \cup T) = f(S) \cup f(T)$.
 - $f(S \cap T) \subseteq f(S) \cap f(T)$.
43. a) Give an example to show that the inclusion in part (b) in Exercise 42 may be proper.
- b) Show that if f is one-to-one, the inclusion in part (b) in Exercise 42 is an equality.
- Let f be a function from the set A to the set B . Let S be a subset of B . We define the **inverse image** of S to be the subset of A whose elements are precisely all preimages of all elements of S . We denote the inverse image of S by $f^{-1}(S)$, so $f^{-1}(S) = \{a \in A \mid f(a) \in S\}$. [Beware: The notation f^{-1} is used in two different ways. Do not confuse the notation introduced here with the notation $f^{-1}(y)$ for the value at y of the inverse of the invertible function f . Notice also that $f^{-1}(S)$, the inverse image of the set S , makes sense for all functions f , not just invertible functions.]
44. Let f be the function from $\mathbf{R}$ to $\mathbf{R}$ defined by $f(x) = x^2$. Find
- $f^{-1}(\{1\})$.
 - $f^{-1}(\{x \mid 0 < x < 1\})$.
 - $f^{-1}(\{x \mid x > 4\})$.
45. Let $g(x) = \lfloor x \rfloor$. Find
- $g^{-1}(\{0\})$.
 - $g^{-1}(\{-1, 0, 1\})$.
 - $g^{-1}(\{x \mid 0 < x < 1\})$.
46. Let f be a function from A to B . Let S and T be subsets of B . Show that
- $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$.
 - $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$.
47. Let f be a function from A to B . Let S be a subset of B . Show that $f^{-1}(\bar{S}) = \overline{f^{-1}(S)}$.
48. Show that $\lfloor x + \frac{1}{2} \rfloor$ is the closest integer to the number x , except when x is midway between two integers, when it is the larger of these two integers.
49. Show that $\lfloor x - \frac{1}{2} \rfloor$ is the closest integer to the number x , except when x is midway between two integers, when it is the smaller of these two integers.
50. Show that if x is a real number, then $\lceil x \rceil - \lfloor x \rfloor = 1$ if x is not an integer and $\lceil x \rceil - \lfloor x \rfloor = 0$ if x is an integer.
51. Show that if x is a real number, then $x - 1 < \lfloor x \rfloor \leq x \leq \lceil x \rceil < x + 1$.
52. Show that if x is a real number and m is an integer, then $\lfloor x + m \rfloor = \lfloor x \rfloor + m$.
53. Show that if x is a real number and n is an integer, then
- $x < n$ if and only if $\lfloor x \rfloor < n$.
 - $n < x$ if and only if $n < \lceil x \rceil$.
54. Show that if x is a real number and n is an integer, then
- $x \leq n$ if and only if $\lfloor x \rfloor \leq n$.
 - $n \leq x$ if and only if $n \leq \lceil x \rceil$.
55. Prove that if n is an integer, then $\lfloor n/2 \rfloor = n/2$ if n is even and $(n-1)/2$ if n is odd.
56. Prove that if x is a real number, then $\lfloor -x \rfloor = -\lceil x \rceil$ and $\lceil -x \rceil = -\lfloor x \rfloor$.
57. The function INT is found on some calculators, where $\text{INT}(x) = \lfloor x \rfloor$ when x is a nonnegative real number and $\text{INT}(x) = \lceil x \rceil$ when x is a negative real number. Show that this INT function satisfies the identity $\text{INT}(-x) = -\text{INT}(x)$.

58. Let a and b be real numbers with $a < b$. Use the floor and/or ceiling functions to express the number of integers n that satisfy the inequality $a \leq n \leq b$.
59. Let a and b be real numbers with $a < b$. Use the floor and/or ceiling functions to express the number of integers n that satisfy the inequality $a < n < b$.
60. How many bytes are required to encode n bits of data where n equals
 a) 4? b) 10? c) 500? d) 3000?
61. How many bytes are required to encode n bits of data where n equals
 a) 7? b) 17? c) 1001? d) 28,800?
62. How many ATM cells (described in Example 30) can be transmitted in 10 seconds over a link operating at the following rates?
 a) 128 kilobits per second (1 kilobit = 1000 bits)
 b) 300 kilobits per second
 c) 1 megabit per second (1 megabit = 1,000,000 bits)
63. Data are transmitted over a particular Ethernet network in blocks of 1500 octets (blocks of 8 bits). How many blocks are required to transmit the following amounts of data over this Ethernet network? (Note that a byte is a synonym for an octet, a kilobyte is 1000 bytes, and a megabyte is 1,000,000 bytes.)
 a) 150 kilobytes of data
 b) 384 kilobytes of data
 c) 1.544 megabytes of data
 d) 45.3 megabytes of data
64. Draw the graph of the function $f(n) = 1 - n^2$ from $\mathbf{Z}$ to $\mathbf{Z}$.
65. Draw the graph of the function $f(x) = \lfloor 2x \rfloor$ from $\mathbf{R}$ to $\mathbf{R}$.
66. Draw the graph of the function $f(x) = \lfloor x/2 \rfloor$ from $\mathbf{R}$ to $\mathbf{R}$.
67. Draw the graph of the function $f(x) = \lfloor x \rfloor + \lfloor x/2 \rfloor$ from $\mathbf{R}$ to $\mathbf{R}$.
68. Draw the graph of the function $f(x) = \lfloor x \rfloor + \lfloor x/2 \rfloor$ from $\mathbf{R}$ to $\mathbf{R}$.
69. Draw graphs of each of these functions.
 a) $f(x) = \lfloor x + \frac{1}{2} \rfloor$ b) $f(x) = \lfloor 2x + 1 \rfloor$
 c) $f(x) = \lfloor x/3 \rfloor$ d) $f(x) = \lceil 1/x \rceil$
 e) $f(x) = \lfloor x - 2 \rfloor + \lfloor x + 2 \rfloor$
 f) $f(x) = \lfloor 2x \rfloor \lfloor x/2 \rfloor$ g) $f(x) = \lceil \lfloor x - \frac{1}{2} \rfloor + \frac{1}{2} \rceil$
70. Draw graphs of each of these functions.
 a) $f(x) = \lfloor 3x - 2 \rfloor$ b) $f(x) = \lfloor 0.2x \rfloor$
 c) $f(x) = \lfloor -1/x \rfloor$ d) $f(x) = \lfloor x^2 \rfloor$
 e) $f(x) = \lfloor x/2 \rfloor \lfloor x/2 \rfloor$ f) $f(x) = \lfloor x/2 \rfloor + \lfloor x/2 \rfloor$
 g) $f(x) = \lfloor 2 \lfloor x/2 \rfloor + \frac{1}{2} \rfloor$
71. Find the inverse function of $f(x) = x^3 + 1$.
72. Suppose that f is an invertible function from Y to Z and g is an invertible function from X to Y . Show that the inverse of the composition $f \circ g$ is given by $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$.
73. Let S be a subset of a universal set U . The **characteristic function** f_S of S is the function from U to the set $\{0, 1\}$ such that $f_S(x) = 1$ if x belongs to S and $f_S(x) = 0$ if x does not belong to S . Let A and B be sets. Show that for all $x \in U$,
 a) $f_{A \cap B}(x) = f_A(x) \cdot f_B(x)$
 b) $f_{A \cup B}(x) = f_A(x) + f_B(x) - f_A(x) \cdot f_B(x)$
 c) $f_{\bar{A}}(x) = 1 - f_A(x)$
 d) $f_{A \oplus B}(x) = f_A(x) + f_B(x) - 2f_A(x)f_B(x)$
74. Suppose that f is a function from A to B , where A and B are finite sets with $|A| = |B|$. Show that f is one-to-one if and only if it is onto.
75. Prove or disprove each of these statements about the floor and ceiling functions.
 a) $\lceil \lfloor x \rfloor \rceil = \lfloor x \rfloor$ for all real numbers x .
 b) $\lfloor 2x \rfloor = 2 \lfloor x \rfloor$ whenever x is a real number.
 c) $\lfloor x \rfloor + \lfloor y \rfloor - \lfloor x + y \rfloor = 0$ or 1 whenever x and y are real numbers.
 d) $\lfloor xy \rfloor = \lfloor x \rfloor \lfloor y \rfloor$ for all real numbers x and y .
 e) $\left\lfloor \frac{x}{2} \right\rfloor = \left\lfloor \frac{x+1}{2} \right\rfloor$ for all real numbers x .
76. Prove or disprove each of these statements about the floor and ceiling functions.
 a) $\lfloor \lceil x \rceil \rfloor = \lfloor x \rfloor$ for all real numbers x .
 b) $\lfloor x + y \rfloor = \lfloor x \rfloor + \lfloor y \rfloor$ for all real numbers x and y .
 c) $\lceil \lfloor x/2 \rfloor / 2 \rceil = \lfloor x/4 \rfloor$ for all real numbers x .
 d) $\lfloor \sqrt{\lfloor x \rfloor} \rfloor = \lfloor \sqrt{x} \rfloor$ for all positive real numbers x .
 e) $\lfloor x \rfloor + \lfloor y \rfloor + \lfloor x + y \rfloor \leq \lfloor 2x \rfloor + \lfloor 2y \rfloor$ for all real numbers x and y .
77. Prove that if x is a positive real number, then
 a) $\lfloor \sqrt{\lfloor x \rfloor} \rfloor = \lfloor \sqrt{x} \rfloor$.
 b) $\lceil \sqrt{\lceil x \rceil} \rceil = \lceil \sqrt{x} \rceil$.
78. Let x be a real number. Show that $\lfloor 3x \rfloor = \lfloor x \rfloor + \lfloor x + \frac{1}{3} \rfloor + \lfloor x + \frac{2}{3} \rfloor$.
79. For each of these partial functions, determine its domain, codomain, domain of definition, and the set of values for which it is undefined. Also, determine whether it is a total function.
 a) $f: \mathbf{Z} \rightarrow \mathbf{R}, f(n) = 1/n$
 b) $f: \mathbf{Z} \rightarrow \mathbf{Z}, f(n) = \lfloor n/2 \rfloor$
 c) $f: \mathbf{Z} \times \mathbf{Z} \rightarrow \mathbf{Q}, f(m, n) = m/n$
 d) $f: \mathbf{Z} \times \mathbf{Z} \rightarrow \mathbf{Z}, f(m, n) = mn$
 e) $f: \mathbf{Z} \times \mathbf{Z} \rightarrow \mathbf{Z}, f(m, n) = m - n$ if $m > n$
80. a) Show that a partial function from A to B can be viewed as a function f^* from A to $B \cup \{u\}$, where u is not an element of B and

$$f^*(a) = \begin{cases} f(a) & \text{if } a \text{ belongs to the domain} \\ & \text{of definition of } f \\ u & \text{if } f \text{ is undefined at } a. \end{cases}$$
 b) Using the construction in (a), find the function f^* corresponding to each partial function in Exercise 79.
81. a) Show that if a set S has cardinality m , where m is a positive integer, then there is a one-to-one correspondence between S and the set $\{1, 2, \dots, m\}$.
 b) Show that if S and T are two sets each with m elements, where m is a positive integer, then there is a one-to-one correspondence between S and T .
- *82. Show that a set S is infinite if and only if there is a proper subset A of S such that there is a one-to-one correspondence between A and S .

from Example 24 we find that

$$\sum_{k=1}^{\infty} kx^{k-1} = \frac{1}{(1-x)^2}.$$

(This differentiation is valid for $|x| < 1$ by a theorem about infinite series.)

Exercises

- Find these terms of the sequence $\{a_n\}$, where $a_n = 2 \cdot (-3)^n + 5^n$.
 a) a_0 b) a_1 c) a_4 d) a_5
- What is the term a_8 of the sequence $\{a_n\}$ if a_n equals
 a) 2^{n-1} ? b) 7 ?
 c) $1 + (-1)^n$? d) $-(-2)^n$?
- What are the terms a_0, a_1, a_2 , and a_3 of the sequence $\{a_n\}$, where a_n equals
 a) $2^n + 1$? b) $(n+1)^{n+1}$?
 c) $\lfloor n/2 \rfloor$? d) $\lfloor n/2 \rfloor + \lceil n/2 \rceil$?
- What are the terms a_0, a_1, a_2 , and a_3 of the sequence $\{a_n\}$, where a_n equals
 a) $(-2)^n$? b) 3 ?
 c) $7 + 4^n$? d) $2^n + (-2)^n$?
- List the first 10 terms of each of these sequences.
 - the sequence that begins with 2 and in which each successive term is 3 more than the preceding term
 - the sequence that lists each positive integer three times, in increasing order
 - the sequence that lists the odd positive integers in increasing order, listing each odd integer twice
 - the sequence whose n th term is $n! - 2^n$
 - the sequence that begins with 3, where each succeeding term is twice the preceding term
 - the sequence whose first term is 2, second term is 4, and each succeeding term is the sum of the two preceding terms
 - the sequence whose n th term is the number of bits in the binary expansion of the number n (defined in Section 4.2)
 - the sequence where the n th term is the number of letters in the English word for the index n
- List the first 10 terms of each of these sequences.
 - the sequence obtained by starting with 10 and obtaining each term by subtracting 3 from the previous term
 - the sequence whose n th term is the sum of the first n positive integers
 - the sequence whose n th term is $3^n - 2^n$
 - the sequence whose n th term is $\lfloor \sqrt{n} \rfloor$
 - the sequence whose first two terms are 1 and 5 and each succeeding term is the sum of the two previous terms
 - the sequence whose n th term is the largest integer whose binary expansion (defined in Section 4.2) has n bits (Write your answer in decimal notation.)
 - the sequence whose terms are constructed sequentially as follows: start with 1, then add 1, then multiply by 1, then add 2, then multiply by 2, and so on
 - the sequence whose n th term is the largest integer k such that $k! \leq n$
- Find at least three different sequences beginning with the terms 1, 2, 4 whose terms are generated by a simple formula or rule.
- Find at least three different sequences beginning with the terms 3, 5, 7 whose terms are generated by a simple formula or rule.
- Find the first five terms of the sequence defined by each of these recurrence relations and initial conditions.
 - $a_n = 6a_{n-1}, a_0 = 2$
 - $a_n = a_{n-1}^2, a_1 = 2$
 - $a_n = a_{n-1} + 3a_{n-2}, a_0 = 1, a_1 = 2$
 - $a_n = na_{n-1} + n^2a_{n-2}, a_0 = 1, a_1 = 1$
 - $a_n = a_{n-1} + a_{n-3}, a_0 = 1, a_1 = 2, a_2 = 0$
- Find the first six terms of the sequence defined by each of these recurrence relations and initial conditions.
 - $a_n = -2a_{n-1}, a_0 = -1$
 - $a_n = a_{n-1} - a_{n-2}, a_0 = 2, a_1 = -1$
 - $a_n = 3a_{n-1}^2, a_0 = 1$
 - $a_n = na_{n-1} + a_{n-2}^2, a_0 = -1, a_1 = 0$
 - $a_n = a_{n-1} - a_{n-2} + a_{n-3}, a_0 = 1, a_1 = 1, a_2 = 2$
- Let $a_n = 2^n + 5 \cdot 3^n$ for $n = 0, 1, 2, \dots$.
 - Find a_0, a_1, a_2, a_3 , and a_4 .
 - Show that $a_2 = 5a_1 - 6a_0, a_3 = 5a_2 - 6a_1$, and $a_4 = 5a_3 - 6a_2$.
 - Show that $a_n = 5a_{n-1} - 6a_{n-2}$ for all integers n with $n \geq 2$.
- Show that the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = -3a_{n-1} + 4a_{n-2}$ if
 - $a_n = 0$. b) $a_n = 1$.
 - $a_n = (-4)^n$. d) $a_n = 2(-4)^n + 3$.
- Is the sequence $\{a_n\}$ a solution of the recurrence relation $a_n = 8a_{n-1} - 16a_{n-2}$ if
 - $a_n = 0$? b) $a_n = 1$?
 - $a_n = 2^n$? d) $a_n = 4^n$?
 - $a_n = n4^n$? f) $a_n = 2 \cdot 4^n + 3n4^n$?
 - $a_n = (-4)^n$? h) $a_n = n^24^n$?
- For each of these sequences find a recurrence relation satisfied by this sequence. (The answers are not unique)

because there are infinitely many different recurrence relations satisfied by any sequence.)

- a) $a_n = 3$ b) $a_n = 2n$
 c) $a_n = 2n + 3$ d) $a_n = 5^n$
 e) $a_n = n^2$ f) $a_n = n^2 + n$
 g) $a_n = n + (-1)^n$ h) $a_n = n!$
15. Show that the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = a_{n-1} + 2a_{n-2} + 2n - 9$ if
- a) $a_n = -n + 2$.
 b) $a_n = 5(-1)^n - n + 2$.
 c) $a_n = 3(-1)^n + 2^n - n + 2$.
 d) $a_n = 7 \cdot 2^n - n + 2$.
16. Find the solution to each of these recurrence relations with the given initial conditions. Use an iterative approach such as that used in Example 10.
- a) $a_n = -a_{n-1}$, $a_0 = 5$
 b) $a_n = a_{n-1} + 3$, $a_0 = 1$
 c) $a_n = a_{n-1} - n$, $a_0 = 4$
 d) $a_n = 2a_{n-1} - 3$, $a_0 = -1$
 e) $a_n = (n + 1)a_{n-1}$, $a_0 = 2$
 f) $a_n = 2na_{n-1}$, $a_0 = 3$
 g) $a_n = -a_{n-1} + n - 1$, $a_0 = 7$
17. Find the solution to each of these recurrence relations and initial conditions. Use an iterative approach such as that used in Example 10.
- a) $a_n = 3a_{n-1}$, $a_0 = 2$
 b) $a_n = a_{n-1} + 2$, $a_0 = 3$
 c) $a_n = a_{n-1} + n$, $a_0 = 1$
 d) $a_n = a_{n-1} + 2n + 3$, $a_0 = 4$
 e) $a_n = 2a_{n-1} - 1$, $a_0 = 1$
 f) $a_n = 3a_{n-1} + 1$, $a_0 = 1$
 g) $a_n = na_{n-1}$, $a_0 = 5$
 h) $a_n = 2na_{n-1}$, $a_0 = 1$
18. A person deposits \$1000 in an account that yields 9% interest compounded annually.
- a) Set up a recurrence relation for the amount in the account at the end of n years.
 b) Find an explicit formula for the amount in the account at the end of n years.
 c) How much money will the account contain after 100 years?
19. Suppose that the number of bacteria in a colony triples every hour.
- a) Set up a recurrence relation for the number of bacteria after n hours have elapsed.
 b) If 100 bacteria are used to begin a new colony, how many bacteria will be in the colony in 10 hours?
20. Assume that the population of the world in 2017 was 7.6 billion and is growing at the rate of 1.12% a year.
- a) Set up a recurrence relation for the population of the world n years after 2017.
 b) Find an explicit formula for the population of the world n years after 2017.
 c) What will the population of the world be in 2050?

21. A factory makes custom sports cars at an increasing rate. In the first month only one car is made, in the second month two cars are made, and so on, with n cars made in the n th month.
- a) Set up a recurrence relation for the number of cars produced in the first n months by this factory.
 b) How many cars are produced in the first year?
 c) Find an explicit formula for the number of cars produced in the first n months by this factory.
22. An employee joined a company in 2017 with a starting salary of \$50,000. Every year this employee receives a raise of \$1000 plus 5% of the salary of the previous year.
- a) Set up a recurrence relation for the salary of this employee n years after 2017.
 b) What will the salary of this employee be in 2025?
 c) Find an explicit formula for the salary of this employee n years after 2017.
23. Find a recurrence relation for the balance $B(k)$ owed at the end of k months on a loan of \$5000 at a rate of 7% if a payment of \$100 is made each month. [Hint: Express $B(k)$ in terms of $B(k - 1)$; the monthly interest is $(0.07/12)B(k - 1)$.]
24. a) Find a recurrence relation for the balance $B(k)$ owed at the end of k months on a loan at a rate of r if a payment P is made on the loan each month. [Hint: Express $B(k)$ in terms of $B(k - 1)$ and note that the monthly interest rate is $r/12$.]
 b) Determine what the monthly payment P should be so that the loan is paid off after T months.
25. For each of these lists of integers, provide a simple formula or rule that generates the terms of an integer sequence that begins with the given list. Assuming that your formula or rule is correct, determine the next three terms of the sequence.
- a) 1, 0, 1, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, ...
 b) 1, 2, 2, 3, 4, 4, 5, 6, 6, 7, 8, 8, ...
 c) 1, 0, 2, 0, 4, 0, 8, 0, 16, 0, ...
 d) 3, 6, 12, 24, 48, 96, 192, ...
 e) 15, 8, 1, -6, -13, -20, -27, ...
 f) 3, 5, 8, 12, 17, 23, 30, 38, 47, ...
 g) 2, 16, 54, 128, 250, 432, 686, ...
 h) 2, 3, 7, 25, 121, 721, 5041, 40321, ...
26. For each of these lists of integers, provide a simple formula or rule that generates the terms of an integer sequence that begins with the given list. Assuming that your formula or rule is correct, determine the next three terms of the sequence.
- a) 3, 6, 11, 18, 27, 38, 51, 66, 83, 102, ...
 b) 7, 11, 15, 19, 23, 27, 31, 35, 39, 43, ...
 c) 1, 10, 11, 100, 101, 110, 111, 1000, 1001, 1010, 1011, ...
 d) 1, 2, 2, 2, 3, 3, 3, 3, 3, 5, 5, 5, 5, 5, ...
 e) 0, 2, 8, 26, 80, 242, 728, 2186, 6560, 19682, ...
 f) 1, 3, 15, 105, 945, 10395, 135135, 2027025, 34459425, ...
 g) 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 1, 1, 1, ...
 h) 2, 4, 16, 256, 65536, 4294967296, ...
- **27. Show that if a_n denotes the n th positive integer that is not a perfect square, then $a_n = n + \{\sqrt{n}\}$, where $\{x\}$ denotes the integer closest to the real number x .



- *28. Let a_n be the n th term of the sequence 1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5, 6, 6, 6, 6, 6, ..., constructed by including the integer k exactly k times. Show that $a_n = \lfloor \sqrt{2n} + \frac{1}{2} \rfloor$.
29. What are the values of these sums?
- a) $\sum_{k=1}^5 (k+1)$ b) $\sum_{j=0}^4 (-2)^j$
- c) $\sum_{i=1}^{10} 3$ d) $\sum_{j=0}^8 (2^{j+1} - 2^j)$
30. What are the values of these sums, where $S = \{1, 3, 5, 7\}$?
- a) $\sum_{j \in S} j$ b) $\sum_{j \in S} j^2$
- c) $\sum_{j \in S} (1/j)$ d) $\sum_{j \in S} 1$
31. What is the value of each of these sums of terms of a geometric progression?
- a) $\sum_{j=0}^8 3 \cdot 2^j$ b) $\sum_{j=1}^8 2^j$
- c) $\sum_{j=2}^8 (-3)^j$ d) $\sum_{j=0}^8 2 \cdot (-3)^j$
32. Find the value of each of these sums.
- a) $\sum_{j=0}^8 (1 + (-1)^j)$ b) $\sum_{j=0}^8 (3^j - 2^j)$
- c) $\sum_{j=0}^8 (2 \cdot 3^j + 3 \cdot 2^j)$ d) $\sum_{j=0}^8 (2^{j+1} - 2^j)$
33. Compute each of these double sums.
- a) $\sum_{i=1}^2 \sum_{j=1}^3 (i+j)$ b) $\sum_{i=0}^2 \sum_{j=0}^3 (2i+3j)$
- c) $\sum_{i=1}^3 \sum_{j=0}^2 i$ d) $\sum_{i=0}^2 \sum_{j=1}^3 ij$
34. Compute each of these double sums.
- a) $\sum_{i=1}^3 \sum_{j=1}^2 (i-j)$ b) $\sum_{i=0}^3 \sum_{j=0}^2 (3i+2j)$
- c) $\sum_{i=1}^3 \sum_{j=0}^2 j$ d) $\sum_{i=0}^2 \sum_{j=0}^3 i^2 j^3$
35. Show that $\sum_{j=1}^n (a_j - a_{j-1}) = a_n - a_0$, where $a_0, a_1, \dots, a_n$ is a sequence of real numbers. This type of sum is called **telescoping**.
36. Use the identity $1/(k(k+1)) = 1/k - 1/(k+1)$ and Exercise 35 to compute $\sum_{k=1}^n 1/(k(k+1))$.
37. Sum both sides of the identity $k^2 - (k-1)^2 = 2k - 1$ from $k = 1$ to $k = n$ and use Exercise 35 to find
- a) a formula for $\sum_{k=1}^n (2k - 1)$ (the sum of the first n odd natural numbers).
- b) a formula for $\sum_{k=1}^n k$.
- *38. Use the technique given in Exercise 35, together with the result of Exercise 37b, to derive the formula for $\sum_{k=1}^n k^2$ given in Table 2. [Hint: Take $a_k = k^3$ in the telescoping sum in Exercise 35.]
39. Find $\sum_{k=100}^{200} k$. (Use Table 2.)
40. Find $\sum_{k=99}^{200} k^3$. (Use Table 2.)
41. Find $\sum_{k=10}^{20} k^2(k-3)$. (Use Table 2.)
42. Find $\sum_{k=10}^{20} (k-1)(2k^2+1)$. (Use Table 2.)
- *43. Find a formula for $\sum_{k=0}^m \lfloor \sqrt{k} \rfloor$, when m is a positive integer.
- *44. Find a formula for $\sum_{k=0}^m \lfloor \sqrt[3]{k} \rfloor$, when m is a positive integer.
- There is also a special notation for products. The product of $a_m, a_{m+1}, \dots, a_n$ is represented by $\prod_{j=m}^n a_j$, read as the product from $j = m$ to $j = n$ of a_j .
45. What are the values of the following products?
- a) $\prod_{i=0}^{10} i$ b) $\prod_{i=5}^8 i$
- c) $\prod_{i=1}^{100} (-1)^i$ d) $\prod_{i=1}^{10} 2$
- Recall that the value of the factorial function at a positive integer n , denoted by $n!$, is the product of the positive integers from 1 to n , inclusive. Also, we specify that $0! = 1$.
46. Express $n!$ using product notation.
47. Find $\sum_{j=0}^4 j!$.
48. Find $\prod_{j=0}^4 j!$.

2.5 Cardinality of Sets

2.5.1 Introduction

In Definition 4 of Section 2.1 we defined the cardinality of a finite set as the number of elements in the set. We use the cardinalities of finite sets to tell us when they have the same size, or when one is bigger than the other. In this section we extend this notion to infinite sets. That is, we will define what it means for two infinite sets to have the same cardinality, providing us with a way to measure the relative sizes of infinite sets.

We will be particularly interested in countably infinite sets, which are sets with the same cardinality as the set of positive integers. We will establish the surprising result that the set of

The continuum hypothesis was stated by Cantor in 1877. He labored unsuccessfully to prove it, becoming extremely dismayed that he could not. By 1900, settling the continuum hypothesis was considered to be among the most important unsolved problems in mathematics. It was the first problem posed by David Hilbert in his 1900 list of open problems in mathematics.

The continuum hypothesis is still an open question and remains an area for active research. However, it has been shown that it can be neither proved nor disproved under the standard set theory axioms in modern mathematics, the Zermelo-Fraenkel axioms. The Zermelo-Fraenkel axioms were formulated to avoid the paradoxes of naive set theory, such as Russell's paradox, but there is much controversy whether they should be replaced by some other set of axioms for set theory.

Exercises

- Determine whether each of these sets is finite, countably infinite, or uncountable. For those that are countably infinite, exhibit a one-to-one correspondence between the set of positive integers and that set.
 - the negative integers
 - the even integers
 - the integers less than 100
 - the real numbers between 0 and $\frac{1}{2}$
 - the positive integers less than 1,000,000,000
 - the integers that are multiples of 7
- Determine whether each of these sets is finite, countably infinite, or uncountable. For those that are countably infinite, exhibit a one-to-one correspondence between the set of positive integers and that set.
 - the integers greater than 10
 - the odd negative integers
 - the integers with absolute value less than 1,000,000
 - the real numbers between 0 and 2
 - the set $A \times \mathbb{Z}^+$ where $A = \{2, 3\}$
 - the integers that are multiples of 10
- Determine whether each of these sets is countable or uncountable. For those that are countably infinite, exhibit a one-to-one correspondence between the set of positive integers and that set.
 - all bit strings not containing the bit 0
 - all positive rational numbers that cannot be written with denominators less than 4
 - the real numbers not containing 0 in their decimal representation
 - the real numbers containing only a finite number of 1s in their decimal representation
- Determine whether each of these sets is countable or uncountable. For those that are countably infinite, exhibit a one-to-one correspondence between the set of positive integers and that set.
 - integers not divisible by 3
 - integers divisible by 5 but not by 7
 - the real numbers with decimal representations consisting of all 1s
 - the real numbers with decimal representations of all 1s or 9s
- Show that a finite group of guests arriving at Hilbert's fully occupied Grand Hotel can be given rooms without evicting any current guest.
- Suppose that Hilbert's Grand Hotel is fully occupied, but the hotel closes all the even numbered rooms for maintenance. Show that all guests can remain in the hotel.
- Suppose that Hilbert's Grand Hotel is fully occupied on the day the hotel expands to a second building which also contains a countably infinite number of rooms. Show that the current guests can be spread out to fill every room of the two buildings of the hotel.
- Show that a countably infinite number of guests arriving at Hilbert's fully occupied Grand Hotel can be given rooms without evicting any current guest.
- *9. Suppose that a countably infinite number of buses, each containing a countably infinite number of guests, arrive at Hilbert's fully occupied Grand Hotel. Show that all the arriving guests can be accommodated without evicting any current guest.
- Give an example of two uncountable sets A and B such that $A - B$ is
 - finite.
 - countably infinite.
 - uncountable.
- Give an example of two uncountable sets A and B such that $A \cap B$ is
 - finite.
 - countably infinite.
 - uncountable.
- Show that if A and B are sets and $A \subset B$ then $|A| \leq |B|$.
- Explain why the set A is countable if and only if $|A| \leq |\mathbb{Z}^+|$.
- Show that if A and B are sets with the same cardinality, then $|A| \leq |B|$ and $|B| \leq |A|$.
- Show that if A and B are sets, A is uncountable, and $A \subseteq B$, then B is uncountable.
- Show that a subset of a countable set is also countable.
- If A is an uncountable set and B is a countable set, must $A - B$ be uncountable?
- Show that if A and B are sets $|A| = |B|$, then $|\mathcal{P}(A)| = |\mathcal{P}(B)|$.
- Show that if A , B , C , and D are sets with $|A| = |B|$ and $|C| = |D|$, then $|A \times C| = |B \times D|$.

20. Show that if $|A| = |B|$ and $|B| = |C|$, then $|A| = |C|$.
21. Show that if A , B , and C are sets such that $|A| \leq |B|$ and $|B| \leq |C|$, then $|A| \leq |C|$.
22. Suppose that A is a countable set. Show that the set B is also countable if there is an onto function f from A to B .
23. Show that if A is an infinite set, then it contains a countably infinite subset.
24. Show that there is no infinite set A such that $|A| < |\mathbf{Z}^+| = \aleph_0$.
25. Prove that if it is possible to label each element of an infinite set S with a finite string of keyboard characters, from a finite list characters, where no two elements of S have the same label, then S is a countably infinite set.
26. Use Exercise 25 to provide a proof different from that in the text that the set of rational numbers is countable. [Hint: Show that you can express a rational number as a string of digits with a slash and possibly a minus sign.]
- *27. Show that the union of a countable number of countable sets is countable.
28. Show that the set $\mathbf{Z}^+ \times \mathbf{Z}^+$ is countable.
- *29. Show that the set of all finite bit strings is countable.
- *30. Show that the set of real numbers that are solutions of quadratic equations $ax^2 + bx + c = 0$, where a , b , and c are integers, is countable.
- *31. Show that $\mathbf{Z}^+ \times \mathbf{Z}^+$ is countable by showing that the polynomial function $f : \mathbf{Z}^+ \times \mathbf{Z}^+ \rightarrow \mathbf{Z}^+$ with $f(m, n) = (m + n - 2)(m + n - 1)/2 + m$ is one-to-one and onto.
- *32. Show that when you substitute $(3n + 1)^2$ for each occurrence of n and $(3m + 1)^2$ for each occurrence of m in the right-hand side of the formula for the function $f(m, n)$ in Exercise 31, you obtain a one-to-one polynomial function $\mathbf{Z} \times \mathbf{Z} \rightarrow \mathbf{Z}$. It is an open question whether there is a one-to-one polynomial function $\mathbf{Q} \times \mathbf{Q} \rightarrow \mathbf{Q}$.
33. Use the Schröder-Bernstein theorem to show that $(0, 1)$ and $[0, 1]$ have the same cardinality.
34. Show that $(0, 1)$ and $\mathbf{R}$ have the same cardinality by
- a) showing that $f(x) = \frac{2x-1}{2x(1-x)}$ is a bijection from $(0, 1)$ to $\mathbf{R}$.
 - b) using the Schröder-Bernstein theorem.
35. Show that there is no one-to-one correspondence from the set of positive integers to the power set of the set of positive integers. [Hint: Assume that there is such a one-to-one correspondence. Represent a subset of the set of positive integers as an infinite bit string with i th bit 1 if i belongs to the subset and 0 otherwise. Suppose that you can list these infinite strings in a sequence indexed by the positive integers. Construct a new bit string with its i th bit equal to the complement of the i th bit of the i th string in the list. Show that this new bit string cannot appear in the list.]
- *36. Show that there is a one-to-one correspondence from the set of subsets of the positive integers to the set real numbers between 0 and 1. Use this result and Exercises 34 and 35 to conclude that $\aleph_0 < |\mathcal{P}(\mathbf{Z}^+)| = |\mathbf{R}|$. [Hint: Look at the first part of the hint for Exercise 35.]
- *37. Show that the set of all computer programs in a particular programming language is countable. [Hint: A computer program written in a programming language can be thought of as a string of symbols from a finite alphabet.]
- *38. Show that the set of functions from the positive integers to the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ is uncountable. [Hint: First set up a one-to-one correspondence between the set of real numbers between 0 and 1 and a subset of these functions. Do this by associating to the real number $0.d_1d_2 \dots d_n \dots$ the function f with $f(n) = d_n$.]
- *39. We say that a function is **computable** if there is a computer program that finds the values of this function. Use Exercises 37 and 38 to show that there are functions that are not computable.
- *40. Show that if S is a set, then there does not exist an onto function f from S to $\mathcal{P}(S)$, the power set of S . Conclude that $|S| < |\mathcal{P}(S)|$. This result is known as **Cantor's theorem**. [Hint: Suppose such a function f existed. Let $T = \{s \in S \mid s \notin f(s)\}$ and show that no element s can exist for which $f(s) = T$.]
- *41. In this exercise, we prove the Schröder-Bernstein theorem. Suppose that A and B are sets where $|A| \leq |B|$ and $|B| \leq |A|$. This means that there are injections $f : A \rightarrow B$ and $g : B \rightarrow A$. To prove the theorem, we must show that there is a bijection $h : A \rightarrow B$, implying that $|A| = |B|$.
- To build h , we construct the *chain* of an element $a \in A$. This chain contains the elements $a, f(a), g(f(a)), f(g(f(a))), g(f(g(f(a))))$, ... It also may contain more elements that precede a , extending the chain backwards. So, if there is a $b \in B$ with $g(b) = a$, then b will be the term of the chain just before a . Because g may not be a surjection, there may not be any such b , so that a is the first element of the chain. If such a b exists, because g is an injection, it is the unique element of B mapped by g to a ; we denote it by $g^{-1}(a)$. (Note that this defines g^{-1} as a partial function from B to A .) We extend the chain backwards as long as possible in the same way, adding $f^{-1}(g^{-1}(a))$, $g^{-1}(f^{-1}(g^{-1}(a)))$, To construct the proof, complete these five parts.
- a) Show that every element in A or in B belongs to exactly one chain.
 - b) Show that there are four types of chains: chains that form a loop, that is, carrying them forward from every element in the chain will eventually return to this element (type 1), chains that go backwards without stopping (type 2), chains that go backwards and end in the set A (type 3), and chains that go backwards and end in the set B (type 4).
 - c) We now define a function $h : A \rightarrow B$. We set $h(a) = f(a)$ when a belongs to a chain of type 1, 2, or 3. Show that we can define $h(a)$ when a is in a chain of type 4, by taking $h(a) = g^{-1}(a)$. In parts (d) and (e), we show that this function is a bijection from A to B , proving the theorem.
 - d) Show that h is one-to-one. (You can consider chains of types 1, 2, and 3 together, but chains of type 4 separately.)
 - e) Show that h is onto. (You need to consider chains of types 1, 2, and 3 together, but chains of type 4 separately.)

We can also define the Boolean powers of a square zero–one matrix. These powers will be used in our subsequent studies of paths in graphs, which are used to model such things as communications paths in computer networks.

Definition 10

Let $\mathbf{A}$ be a square zero–one matrix and let r be a positive integer. The r th *Boolean power* of $\mathbf{A}$ is the Boolean product of r factors of $\mathbf{A}$. The r th Boolean product of $\mathbf{A}$ is denoted by $\mathbf{A}^{[r]}$. Hence,

$$\mathbf{A}^{[r]} = \underbrace{\mathbf{A} \odot \mathbf{A} \odot \mathbf{A} \odot \cdots \odot \mathbf{A}}_{r \text{ times}}.$$

(This is well defined because the Boolean product of matrices is associative.) We also define $\mathbf{A}^{[0]}$ to be $\mathbf{I}_n$.

EXAMPLE 9 Let $\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$. Find $\mathbf{A}^{[n]}$ for all positive integers n .

Solution: We find that

$$\mathbf{A}^{[2]} = \mathbf{A} \odot \mathbf{A} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

We also find that

$$\mathbf{A}^{[3]} = \mathbf{A}^{[2]} \odot \mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \quad \mathbf{A}^{[4]} = \mathbf{A}^{[3]} \odot \mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Additional computation shows that

$$\mathbf{A}^{[5]} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

The reader can now see that $\mathbf{A}^{[n]} = \mathbf{A}^{[5]}$ for all positive integers n with $n \geq 5$. ◀

Exercises

1. Let $\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 & 3 \\ 2 & 0 & 4 & 6 \\ 1 & 1 & 3 & 7 \end{bmatrix}$.

- What size is $\mathbf{A}$?
- What is the third column of $\mathbf{A}$?
- What is the second row of $\mathbf{A}$?
- What is the element of $\mathbf{A}$ in the (3, 2)th position?
- What is $\mathbf{A}^t$?

2. Find $\mathbf{A} + \mathbf{B}$, where

a) $\mathbf{A} = \begin{bmatrix} 1 & 0 & 4 \\ -1 & 2 & 2 \\ 0 & -2 & -3 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} -1 & 3 & 5 \\ 2 & 2 & -3 \\ 2 & -3 & 0 \end{bmatrix}$.

b) $\mathbf{A} = \begin{bmatrix} -1 & 0 & 5 & 6 \\ -4 & -3 & 5 & -2 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} -3 & 9 & -3 & 4 \\ 0 & -2 & -1 & 2 \end{bmatrix}$.

3. Find $\mathbf{AB}$ if

a) $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 0 & 4 \\ 1 & 3 \end{bmatrix}$.

$$\text{b) } \mathbf{A} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 2 & 3 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 3 & -2 & -1 \\ 1 & 0 & 2 \end{bmatrix}.$$

$$\text{c) } \mathbf{A} = \begin{bmatrix} 4 & -3 \\ 3 & -1 \\ 0 & -2 \\ -1 & 5 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} -1 & 3 & 2 & -2 \\ 0 & -1 & 4 & -3 \end{bmatrix}.$$

4. Find the product $\mathbf{AB}$, where

$$\text{a) } \mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ -1 & 1 & 0 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 0 & 1 & -1 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix}.$$

$$\text{b) } \mathbf{A} = \begin{bmatrix} 1 & -3 & 0 \\ 1 & 2 & 2 \\ 2 & 1 & -1 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 1 & -1 & 2 & 3 \\ -1 & 0 & 3 & -1 \\ -3 & -2 & 0 & 2 \end{bmatrix}.$$

$$\text{c) } \mathbf{A} = \begin{bmatrix} 0 & -1 \\ 7 & 2 \\ -4 & -3 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 4 & -1 & 2 & 3 & 0 \\ -2 & 0 & 3 & 4 & 1 \end{bmatrix}.$$

5. Find a matrix $\mathbf{A}$ such that

$$\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} \mathbf{A} = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix}.$$

[Hint: Finding $\mathbf{A}$ requires that you solve systems of linear equations.]

6. Find a matrix $\mathbf{A}$ such that

$$\begin{bmatrix} 1 & 3 & 2 \\ 2 & 1 & 1 \\ 4 & 0 & 3 \end{bmatrix} \mathbf{A} = \begin{bmatrix} 7 & 1 & 3 \\ 1 & 0 & 3 \\ -1 & -3 & 7 \end{bmatrix}.$$

7. Let $\mathbf{A}$ be an $m \times n$ matrix and let $\mathbf{0}$ be the $m \times n$ matrix that has all entries equal to zero. Show that $\mathbf{A} = \mathbf{0} + \mathbf{A} = \mathbf{A} + \mathbf{0}$.

8. Show that matrix addition is commutative; that is, show that if $\mathbf{A}$ and $\mathbf{B}$ are both $m \times n$ matrices, then $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$.

9. Show that matrix addition is associative; that is, show that if $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ are all $m \times n$ matrices, then $\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C}$.

10. Let $\mathbf{A}$ be a 3×4 matrix, $\mathbf{B}$ be a 4×5 matrix, and $\mathbf{C}$ be a 4×4 matrix. Determine which of the following products are defined and find the size of those that are defined.

- a) $\mathbf{AB}$ b) $\mathbf{BA}$ c) $\mathbf{AC}$
d) $\mathbf{CA}$ e) $\mathbf{BC}$ f) $\mathbf{CB}$

11. What do we know about the sizes of the matrices $\mathbf{A}$ and $\mathbf{B}$ if both of the products $\mathbf{AB}$ and $\mathbf{BA}$ are defined?

12. In this exercise we show that matrix multiplication is distributive over matrix addition.

- a) Suppose that $\mathbf{A}$ and $\mathbf{B}$ are $m \times k$ matrices and that $\mathbf{C}$ is a $k \times n$ matrix. Show that $(\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}$.
b) Suppose that $\mathbf{C}$ is an $m \times k$ matrix and that $\mathbf{A}$ and $\mathbf{B}$ are $k \times n$ matrices. Show that $\mathbf{C}(\mathbf{A} + \mathbf{B}) = \mathbf{CA} + \mathbf{CB}$.

13. In this exercise we show that matrix multiplication is associative. Suppose that $\mathbf{A}$ is an $m \times p$ matrix, $\mathbf{B}$ is a $p \times k$ matrix, and $\mathbf{C}$ is a $k \times n$ matrix. Show that $\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}$.

14. The $n \times n$ matrix $\mathbf{A} = [a_{ij}]$ is called a **diagonal matrix** if $a_{ij} = 0$ when $i \neq j$. Show that the product of two $n \times n$

diagonal matrices is again a diagonal matrix. Give a simple rule for determining this product.

15. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Find a formula for $\mathbf{A}^n$, whenever n is a positive integer.

16. Show that $(\mathbf{A}')' = \mathbf{A}$.

17. Let $\mathbf{A}$ and $\mathbf{B}$ be two $n \times n$ matrices. Show that

$$\text{a) } (\mathbf{A} + \mathbf{B})' = \mathbf{A}' + \mathbf{B}'.$$

$$\text{b) } (\mathbf{AB})' = \mathbf{B}'\mathbf{A}'.$$

If $\mathbf{A}$ and $\mathbf{B}$ are $n \times n$ matrices with $\mathbf{AB} = \mathbf{BA} = \mathbf{I}_n$, then $\mathbf{B}$ is called the **inverse** of $\mathbf{A}$ (this terminology is appropriate because such a matrix $\mathbf{B}$ is unique) and $\mathbf{A}$ is said to be **invertible**. The notation $\mathbf{B} = \mathbf{A}^{-1}$ denotes that $\mathbf{B}$ is the inverse of $\mathbf{A}$.

18. Show that

$$\begin{bmatrix} 2 & 3 & -1 \\ 1 & 2 & 1 \\ -1 & -1 & 3 \end{bmatrix}$$

is the inverse of

$$\begin{bmatrix} 7 & -8 & 5 \\ -4 & 5 & -3 \\ 1 & -1 & 1 \end{bmatrix}.$$

19. Let $\mathbf{A}$ be the 2×2 matrix

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Show that if $ad - bc \neq 0$, then

$$\mathbf{A}^{-1} = \begin{bmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix}.$$

20. Let

$$\mathbf{A} = \begin{bmatrix} -1 & 2 \\ 1 & 3 \end{bmatrix}.$$

a) Find $\mathbf{A}^{-1}$. [Hint: Use Exercise 19.]

b) Find $\mathbf{A}^3$.

c) Find $(\mathbf{A}^{-1})^3$.

d) Use your answers to (b) and (c) to show that $(\mathbf{A}^{-1})^3$ is the inverse of $\mathbf{A}^3$.

21. Let $\mathbf{A}$ be an invertible matrix. Show that $(\mathbf{A}^n)^{-1} = (\mathbf{A}^{-1})^n$ whenever n is a positive integer.

22. Let $\mathbf{A}$ be a matrix. Show that the matrix $\mathbf{AA}^t$ is symmetric. [Hint: Show that this matrix equals its transpose with the help of Exercise 17b.]

23. Suppose that $\mathbf{A}$ is an $n \times n$ matrix where n is a positive integer. Show that $\mathbf{A} + \mathbf{A}^t$ is symmetric.

24. a) Show that the system of simultaneous linear equations

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = b_n.$$

in the variables $x_1, x_2, \dots, x_n$ can be expressed as $\mathbf{AX} = \mathbf{B}$, where $\mathbf{A} = [a_{ij}]$, $\mathbf{X}$ is an $n \times 1$ matrix with x_i the entry in its i th row, and $\mathbf{B}$ is an $n \times 1$ matrix with b_i the entry in its i th row.

b) Show that if the matrix $\mathbf{A} = [a_{ij}]$ is invertible (as defined in the preamble to Exercise 18), then the solution of the system in part (a) can be found using the equation $\mathbf{X} = \mathbf{A}^{-1}\mathbf{B}$.

25. Use Exercises 18 and 24 to solve the system

$$7x_1 - 8x_2 + 5x_3 = 5$$

$$-4x_1 + 5x_2 - 3x_3 = -3$$

$$x_1 - x_2 + x_3 = 0$$

26. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Find

a) $\mathbf{A} \vee \mathbf{B}$. b) $\mathbf{A} \wedge \mathbf{B}$. c) $\mathbf{A} \odot \mathbf{B}$.

27. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

Find

a) $\mathbf{A} \vee \mathbf{B}$. b) $\mathbf{A} \wedge \mathbf{B}$. c) $\mathbf{A} \odot \mathbf{B}$.

28. Find the Boolean product of $\mathbf{A}$ and $\mathbf{B}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}.$$

29. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

Find

a) $\mathbf{A}^{[2]}$. b) $\mathbf{A}^{[3]}$.

c) $\mathbf{A} \vee \mathbf{A}^{[2]} \vee \mathbf{A}^{[3]}$.

30. Let $\mathbf{A}$ be a zero-one matrix. Show that

a) $\mathbf{A} \vee \mathbf{A} = \mathbf{A}$. b) $\mathbf{A} \wedge \mathbf{A} = \mathbf{A}$.

31. In this exercise we show that the meet and join operations are commutative. Let $\mathbf{A}$ and $\mathbf{B}$ be $m \times n$ zero-one matrices. Show that

a) $\mathbf{A} \vee \mathbf{B} = \mathbf{B} \vee \mathbf{A}$. b) $\mathbf{B} \wedge \mathbf{A} = \mathbf{A} \wedge \mathbf{B}$.

32. In this exercise we show that the meet and join operations are associative. Let $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ be $m \times n$ zero-one matrices. Show that

a) $(\mathbf{A} \vee \mathbf{B}) \vee \mathbf{C} = \mathbf{A} \vee (\mathbf{B} \vee \mathbf{C})$.

b) $(\mathbf{A} \wedge \mathbf{B}) \wedge \mathbf{C} = \mathbf{A} \wedge (\mathbf{B} \wedge \mathbf{C})$.

33. We will establish distributive laws of the meet over the join operation in this exercise. Let $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ be $m \times n$ zero-one matrices. Show that

a) $\mathbf{A} \vee (\mathbf{B} \wedge \mathbf{C}) = (\mathbf{A} \vee \mathbf{B}) \wedge (\mathbf{A} \vee \mathbf{C})$.

b) $\mathbf{A} \wedge (\mathbf{B} \vee \mathbf{C}) = (\mathbf{A} \wedge \mathbf{B}) \vee (\mathbf{A} \wedge \mathbf{C})$.

34. Let $\mathbf{A}$ be an $n \times n$ zero-one matrix. Let $\mathbf{I}$ be the $n \times n$ identity matrix. Show that $\mathbf{A} \odot \mathbf{I} = \mathbf{I} \odot \mathbf{A} = \mathbf{A}$.

35. In this exercise we will show that the Boolean product of zero-one matrices is associative. Assume that $\mathbf{A}$ is an $m \times p$ zero-one matrix, $\mathbf{B}$ is a $p \times k$ zero-one matrix, and $\mathbf{C}$ is a $k \times n$ zero-one matrix. Show that $\mathbf{A} \odot (\mathbf{B} \odot \mathbf{C}) = (\mathbf{A} \odot \mathbf{B}) \odot \mathbf{C}$.

Key Terms and Results

TERMS

set: an unordered collection of distinct objects

axiom: a basic assumption of a theory

paradox: a logical inconsistency

element, member of a set: an object in a set

roster method: a method that describes a set by listing its elements

set builder notation: the notation that describes a set by stating a property an element must have to be a member

multiset: an unordered collection of objects where objects can occur multiple times

$\emptyset$ (empty set, null set): the set with no members

universal set: the set containing all objects under consideration

Venn diagram: a graphical representation of a set or sets

$S = T$ (set equality): S and T have the same elements

$S \subseteq T$ (S is a subset of T): every element of S is also an element of T

$S \subset T$ (S is a proper subset of T): S is a subset of T and $S \neq T$

finite set: a set with n elements, where n is a nonnegative integer

infinite set: a set that is not finite

$|S|$ (the cardinality of S): the number of elements in S

$P(S)$ (the power set of S): the set of all subsets of S

$A \cup B$ (the union of A and B): the set containing those elements that are in at least one of A and B

$A \cap B$ (the intersection of A and B): the set containing those elements that are in both A and B .

$A - B$ (the difference of A and B): the set containing those elements that are in A but not in B

$\bar{A}$ (the complement of A): the set of elements in the universal set that are not in A

$A \oplus B$ (the symmetric difference of A and B): the set containing those elements in exactly one of A and B

membership table: a table displaying the membership of elements in sets

function from A to B : an assignment of exactly one element of B to each element of A

domain of f : the set A , where f is a function from A to B

codomain of f : the set B , where f is a function from A to B

b is the image of a under f : $b = f(a)$

a is a preimage of b under f : $f(a) = b$

range of f : the set of images of f

onto function, surjection: a function from A to B such that every element of B is the image of some element in A

one-to-one function, injection: a function such that the images of elements in its domain are distinct

one-to-one correspondence, bijection: a function that is both one-to-one and onto

inverse of f : the function that reverses the correspondence given by f (when f is a bijection)

$f \circ g$ (composition of f and g): the function that assigns $f(g(x))$ to x

$\lfloor x \rfloor$ (floor function): the largest integer not exceeding x

$\lceil x \rceil$ (ceiling function): the smallest integer greater than or equal to x

partial function: an assignment to each element in a subset of the domain a unique element in the codomain

sequence: a function with domain that is a subset of the set of integers

geometric progression: a sequence of the form $a, ar, ar^2, \dots$, where a and r are real numbers

arithmetic progression: a sequence of the form $a, a + d, a + 2d, \dots$, where a and d are real numbers

string: a finite sequence

empty string: a string of length zero

recurrence relation: a equation that expresses the n th term a_n of a sequence in terms of one or more of the previous terms of the sequence for all integers n greater than a particular integer

$\sum_{i=1}^n a_i$: the sum $a_1 + a_2 + \dots + a_n$

$\prod_{i=1}^n a_i$: the product $a_1 a_2 \dots a_n$

cardinality: two sets A and B have the same cardinality if there is a one-to-one correspondence from A to B

countable set: a set that either is finite or can be placed in one-to-one correspondence with the set of positive integers

uncountable set: a set that is not countable

$\aleph_0$ (aleph null): the cardinality of a countable set

c : the cardinality of the set of real numbers

Cantor diagonalization argument: a proof technique used to show that the set of real numbers is uncountable

computable function: a function for which there is a computer program in some programming language that finds its values

uncomputable function: a function for which no computer program in a programming language exists that finds its values

continuum hypothesis: the statement that no set A exists such that $\aleph_0 < |A| < c$

matrix: a rectangular array of numbers

matrix addition: see page 188

matrix multiplication: see page 189

I_n (identity matrix of order n): the $n \times n$ matrix that has entries equal to 1 on its diagonal and 0s elsewhere

A^t (transpose of A): the matrix obtained from A by interchanging the rows and columns

symmetric matrix: a matrix is symmetric if it equals its transpose

zero-one matrix: a matrix with each entry equal to either 0 or 1

$A \vee B$ (the join of A and B): see page 191

$A \wedge B$ (the meet of A and B): see page 191

$A \odot B$ (the Boolean product of A and B): see page 192

RESULTS

The set identities given in Table 1 in Section 2.2

The summation formulae in Table 2 in Section 2.4

The set of rational numbers is countable.

The set of real numbers is uncountable.

Review Questions

1. Explain what it means for one set to be a subset of another set. How do you prove that one set is a subset of another set?
2. What is the empty set? Show that the empty set is a subset of every set.
3. a) Define $|S|$, the cardinality of the set S .
b) Give a formula for $|A \cup B|$, where A and B are sets.
4. a) Define the power set of a set S .
b) When is the empty set in the power set of a set S ?
c) How many elements does the power set of a set S with n elements have?

5. a) Define the union, intersection, difference, and symmetric difference of two sets.
b) What are the union, intersection, difference, and symmetric difference of the set of positive integers and the set of odd integers?
6. a) Explain what it means for two sets to be equal.
b) Describe as many of the ways as you can to show that two sets are equal.
c) Show in at least two different ways that the sets $A - (B \cap C)$ and $(A - B) \cup (A - C)$ are equal.
7. Explain the relationship between logical equivalences and set identities.
8. a) Define the domain, codomain, and range of a function.
b) Let $f(n)$ be the function from the set of integers to the set of integers such that $f(n) = n^2 + 1$. What are the domain, codomain, and range of this function?
9. a) Define what it means for a function from the set of positive integers to the set of positive integers to be one-to-one.
b) Define what it means for a function from the set of positive integers to the set of positive integers to be onto.
c) Give an example of a function from the set of positive integers to the set of positive integers that is both one-to-one and onto.
d) Give an example of a function from the set of positive integers to the set of positive integers that is one-to-one but not onto.
- e) Give an example of a function from the set of positive integers to the set of positive integers that is not one-to-one but is onto.
- f) Give an example of a function from the set of positive integers to the set of positive integers that is neither one-to-one nor onto.
10. a) Define the inverse of a function.
b) When does a function have an inverse?
c) Does the function $f(n) = 10 - n$ from the set of integers to the set of integers have an inverse? If so, what is it?
11. a) Define the floor and ceiling functions from the set of real numbers to the set of integers.
b) For which real numbers x is it true that $\lfloor x \rfloor = \lceil x \rceil$?
12. Conjecture a formula for the terms of the sequence that begins 8, 14, 32, 86, 248 and find the next three terms of your sequence.
13. Suppose that $a_n = a_{n-1} - 5$ for $n = 1, 2, \dots$. Find a formula for a_n .
14. What is the sum of the terms of the geometric progression $a + ar + \dots + ar^n$ when $r \neq 1$?
15. Show that the set of odd integers is countable.
16. Give an example of an uncountable set.
17. Define the product of two matrices **A** and **B**. When is this product defined?
18. Show that matrix multiplication is not commutative.

Supplementary Exercises

1. Let A be the set of English words that contain the letter x , and let B be the set of English words that contain the letter q . Express each of these sets as a combination of A and B .
a) The set of English words that do not contain the letter x .
b) The set of English words that contain both an x and a q .
c) The set of English words that contain an x but not a q .
d) The set of English words that do not contain either an x or a q .
e) The set of English words that contain an x or a q , but not both.
2. Show that if A is a subset of B , then the power set of A is a subset of the power set of B .
3. Suppose that A and B are sets such that the power set of A is a subset of the power set of B . Does it follow that A is a subset of B ?
4. Let **E** denote the set of even integers and **O** denote the set of odd integers. As usual, let **Z** denote the set of all integers. Determine each of these sets.
a) $\mathbf{E} \cup \mathbf{O}$ b) $\mathbf{E} \cap \mathbf{O}$ c) $\mathbf{Z} - \mathbf{E}$ d) $\mathbf{Z} - \mathbf{O}$
5. Show that if A and B are sets, then $A - (A - B) = A \cap B$.
6. Let A and B be sets. Show that $A \subseteq B$ if and only if $A \cap B = A$.
7. Let A , B , and C be sets. Show that $(A - B) - C$ is not necessarily equal to $A - (B - C)$.
8. Suppose that A , B , and C are sets. Prove or disprove that $(A - B) - C = (A - C) - B$.
9. Suppose that A , B , C , and D are sets. Prove or disprove that $(A - B) - (C - D) = (A - C) - (B - D)$.
10. Show that if A and B are finite sets, then $|A \cap B| \leq |A \cup B|$. Determine when this relationship is an equality.
11. Let A and B be sets in a finite universal set U . List the following in order of increasing size.
a) $|A|$, $|A \cup B|$, $|A \cap B|$, $|U|$, $|\emptyset|$
b) $|A - B|$, $|A \oplus B|$, $|A| + |B|$, $|A \cup B|$, $|\emptyset|$
12. Let A and B be subsets of the finite universal set U . Show that $|\overline{A \cap B}| = |U| - |A| - |B| + |A \cap B|$.
13. Let f and g be functions from $\{1, 2, 3, 4\}$ to $\{a, b, c, d\}$ and from $\{a, b, c, d\}$ to $\{1, 2, 3, 4\}$, respectively, with $f(1) = d$, $f(2) = c$, $f(3) = a$, and $f(4) = b$, and $g(a) = 2$, $g(b) = 1$, $g(c) = 3$, and $g(d) = 2$.
a) Is f one-to-one? Is g one-to-one?
b) Is f onto? Is g onto?
c) Does either f or g have an inverse? If so, find this inverse.

14. Suppose that f is a function from A to B where A and B are finite sets. Explain why $|f(S)| \leq |S|$ for all subsets S of A .
15. Suppose that f is a function from A to B where A and B are finite sets. Explain why $|f(S)| = |S|$ for all subsets S of A if and only if f is one-to-one.

Suppose that f is a function from A to B . We define the function S_f from $\mathcal{P}(A)$ to $\mathcal{P}(B)$ by the rule $S_f(X) = f(X)$ for each subset X of A . Similarly, we define the function $S_{f^{-1}}$ from $\mathcal{P}(B)$ to $\mathcal{P}(A)$ by the rule $S_{f^{-1}}(Y) = f^{-1}(Y)$ for each subset Y of B . Here, we are using Definition 4, and the definition of the inverse image of a set found in the preamble to Exercise 44, both in Section 2.3.

- *16. Suppose that f is a function from the set A to the set B . Prove that
- if f is one-to-one, then S_f is a one-to-one function from $\mathcal{P}(A)$ to $\mathcal{P}(B)$.
 - if f is onto function, then S_f is an onto function from $\mathcal{P}(A)$ to $\mathcal{P}(B)$.
 - if f is onto function, then $S_{f^{-1}}$ is a one-to-one function from $\mathcal{P}(B)$ to $\mathcal{P}(A)$.
 - if f is one-to-one, then $S_{f^{-1}}$ is an onto function from $\mathcal{P}(B)$ to $\mathcal{P}(A)$.
 - if f is a one-to-one correspondence, then S_f is a one-to-one correspondence from $\mathcal{P}(A)$ to $\mathcal{P}(B)$ and $S_{f^{-1}}$ is a one-to-one correspondence from $\mathcal{P}(B)$ to $\mathcal{P}(A)$. [Hint: Use parts (a)–(d).]
17. Prove that if f and g are functions from A to B and $S_f = S_g$ (using the definition in the preamble to Exercise 16), then $f(x) = g(x)$ for all $x \in A$.
18. Show that if n is an integer, then $n = \lceil n/2 \rceil + \lfloor n/2 \rfloor$.
19. For which real numbers x and y is it true that $\lfloor x + y \rfloor = \lfloor x \rfloor + \lfloor y \rfloor$?
20. For which real numbers x and y is it true that $\lceil x + y \rceil = \lceil x \rceil + \lceil y \rceil$?
21. For which real numbers x and y is it true that $\lceil x + y \rceil = \lceil x \rceil + \lfloor y \rfloor$?
22. Prove that $\lfloor n/2 \rfloor \lceil n/2 \rceil = \lfloor n^2/4 \rfloor$ for all integers n .
23. Prove that if m is an integer, then $\lfloor x \rfloor + \lfloor m - x \rfloor = m - 1$, unless x is an integer, in which case, it equals m .
24. Prove that if x is a real number, then $\lfloor \lfloor x/2 \rfloor / 2 \rfloor = \lfloor x/4 \rfloor$.
25. Prove that if n is an odd integer, then $\lfloor n^2/4 \rfloor = (n^2 + 3)/4$.
26. Prove that if m and n are positive integers and x is a real number, then

$$\left\lfloor \frac{\lfloor x \rfloor + n}{m} \right\rfloor = \left\lfloor \frac{x + n}{m} \right\rfloor.$$

- *27. Prove that if m is a positive integer and x is a real number, then

$$\begin{aligned} \lfloor mx \rfloor &= \lfloor x \rfloor + \left\lfloor x + \frac{1}{m} \right\rfloor + \left\lfloor x + \frac{2}{m} \right\rfloor + \cdots \\ &\quad + \left\lfloor x + \frac{m-1}{m} \right\rfloor. \end{aligned}$$

- *28. We define the **Ulam numbers** by setting $u_1 = 1$ and $u_2 = 2$. Furthermore, after determining whether the integers less than n are Ulam numbers, we set n equal to the next Ulam number if it can be written uniquely as the sum of two different Ulam numbers. Note that $u_3 = 3$, $u_4 = 4$, $u_5 = 6$, and $u_6 = 8$.
- Find the first 20 Ulam numbers.
 - Prove that there are infinitely many Ulam numbers.
29. Determine the value of $\prod_{k=1}^{100} \frac{k+1}{k}$. (The notation used here for products is defined in the preamble to Exercise 45 in Section 2.4.)
- *30. Determine a rule for generating the terms of the sequence that begins 1, 3, 4, 8, 15, 27, 50, 92, ..., and find the next four terms of the sequence.
- *31. Determine a rule for generating the terms of the sequence that begins 2, 3, 3, 5, 10, 13, 39, 43, 172, 177, 885, 891, ..., and find the next four terms of the sequence.
32. Show that the set of irrational numbers is an uncountable set.
33. Show that the set S is a countable set if there is a function f from S to the positive integers such that $f^{-1}(j)$ is countable whenever j is a positive integer.
34. Show that the set of all finite subsets of the set of positive integers is a countable set.
- **35. Show that $|\mathbf{R} \times \mathbf{R}| = |\mathbf{R}|$. [Hint: Use the Schröder-Bernstein theorem to show that $|(0, 1) \times (0, 1)| = |(0, 1)|$. To construct an injection from $(0, 1) \times (0, 1)$ to $(0, 1)$, suppose that $(x, y) \in (0, 1) \times (0, 1)$. Map (x, y) to the number with decimal expansion formed by alternating between the digits in the decimal expansions of x and y , which do not end with an infinite string of 9s.]
- **36. Show that $\mathbf{C}$, the set of complex numbers, has the same cardinality as $\mathbf{R}$, the set of real numbers.
37. Find $\mathbf{A}^n$ if $\mathbf{A}$ is

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

38. Show that if $\mathbf{A} = c\mathbf{I}$, where c is a real number and $\mathbf{I}$ is the $n \times n$ identity matrix, then $\mathbf{AB} = \mathbf{BA}$ whenever $\mathbf{B}$ is an $n \times n$ matrix.
39. Show that if $\mathbf{A}$ is a 2×2 matrix such that $\mathbf{AB} = \mathbf{BA}$ whenever $\mathbf{B}$ is a 2×2 matrix, then $\mathbf{A} = c\mathbf{I}$, where c is a real number and $\mathbf{I}$ is the 2×2 identity matrix.
40. Show that if $\mathbf{A}$ and $\mathbf{B}$ are invertible matrices and $\mathbf{AB}$ exists, then $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$.
41. Let $\mathbf{A}$ be an $n \times n$ matrix and let $\mathbf{0}$ be the $n \times n$ matrix all of whose entries are zero. Show that the following are true.
- $\mathbf{A} \odot \mathbf{0} = \mathbf{0} \odot \mathbf{A} = \mathbf{0}$
 - $\mathbf{A} \vee \mathbf{0} = \mathbf{0} \vee \mathbf{A} = \mathbf{A}$
 - $\mathbf{A} \wedge \mathbf{0} = \mathbf{0} \wedge \mathbf{A} = \mathbf{0}$